

Rookery Bay National Estuarine Research Reserve

SEACAR Water Quality Analysis

Last compiled on 15 July, 2026

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Indicators

Nutrients

Total Nitrogen - Discrete

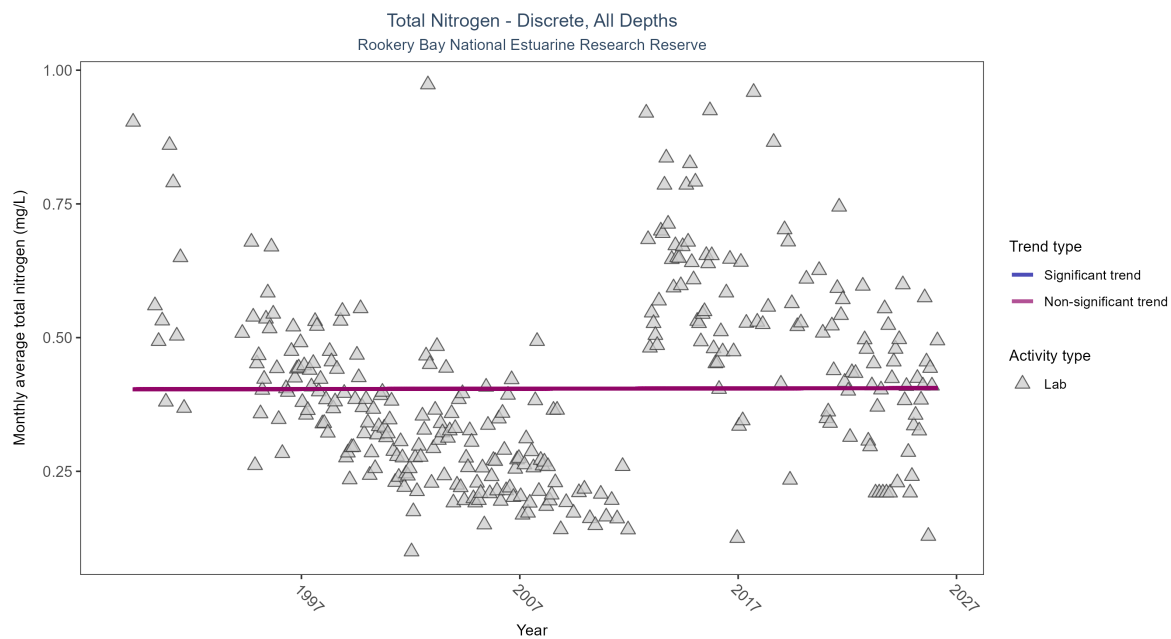


Figure 1: Scatter plot of monthly average total nitrogen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only nitrogen values obtained from laboratory analyses (triangles) are included in the plot.

Table 1: Total nitrogen showed no detectable trend between 1989 and 2026.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	4758	36	1989 - 2026	0.31975	0.00493	0.40336	0.00006	0.9484

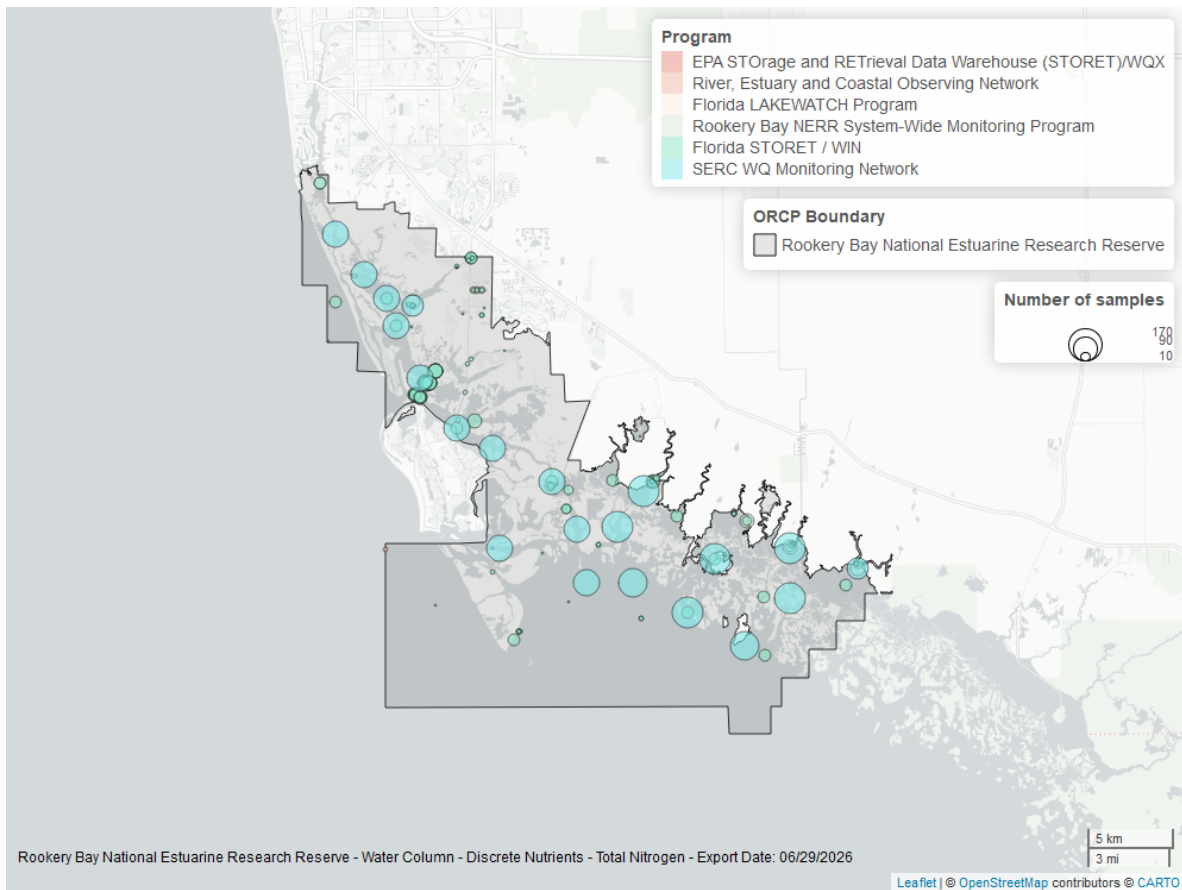


Figure 2: Map showing location of discrete water quality sampling locations within the boundaries of *Rookery Bay National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Total Phosphorus - Discrete

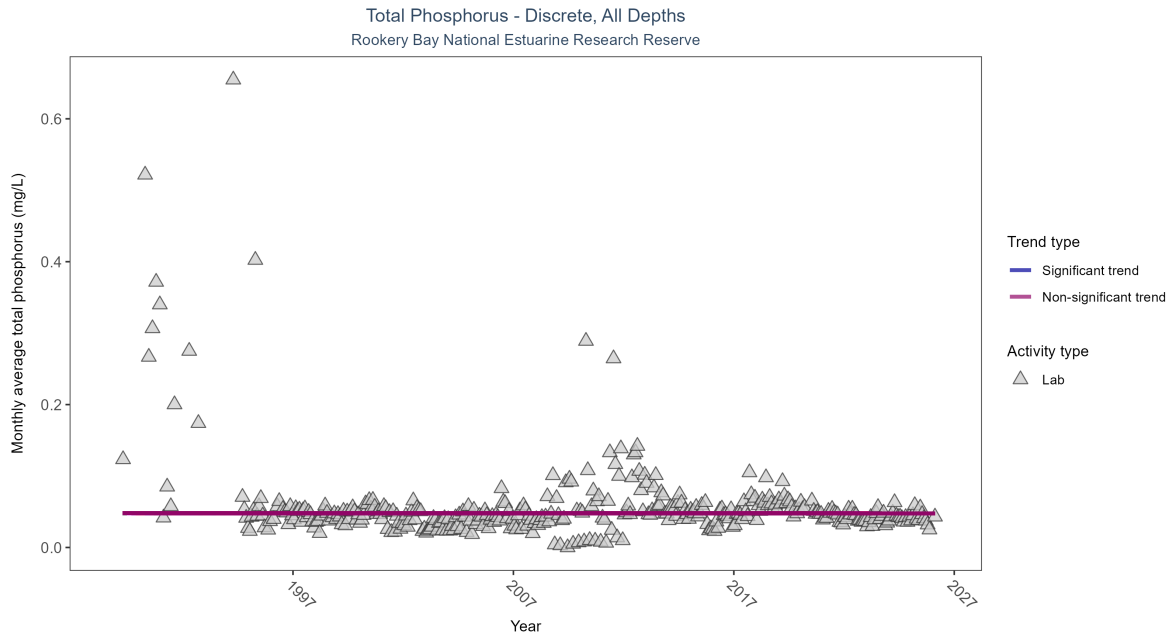


Figure 3: Scatter plot of monthly average total phosphorus over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only phosphorus values obtained from laboratory analyses (triangles) are included in the plot.

Table 2: Total phosphorus showed no detectable trend between 1989 and 2026.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	4913	37	1989 - 2026	0.04	0.00069	0.04807	-0.00001	0.9396

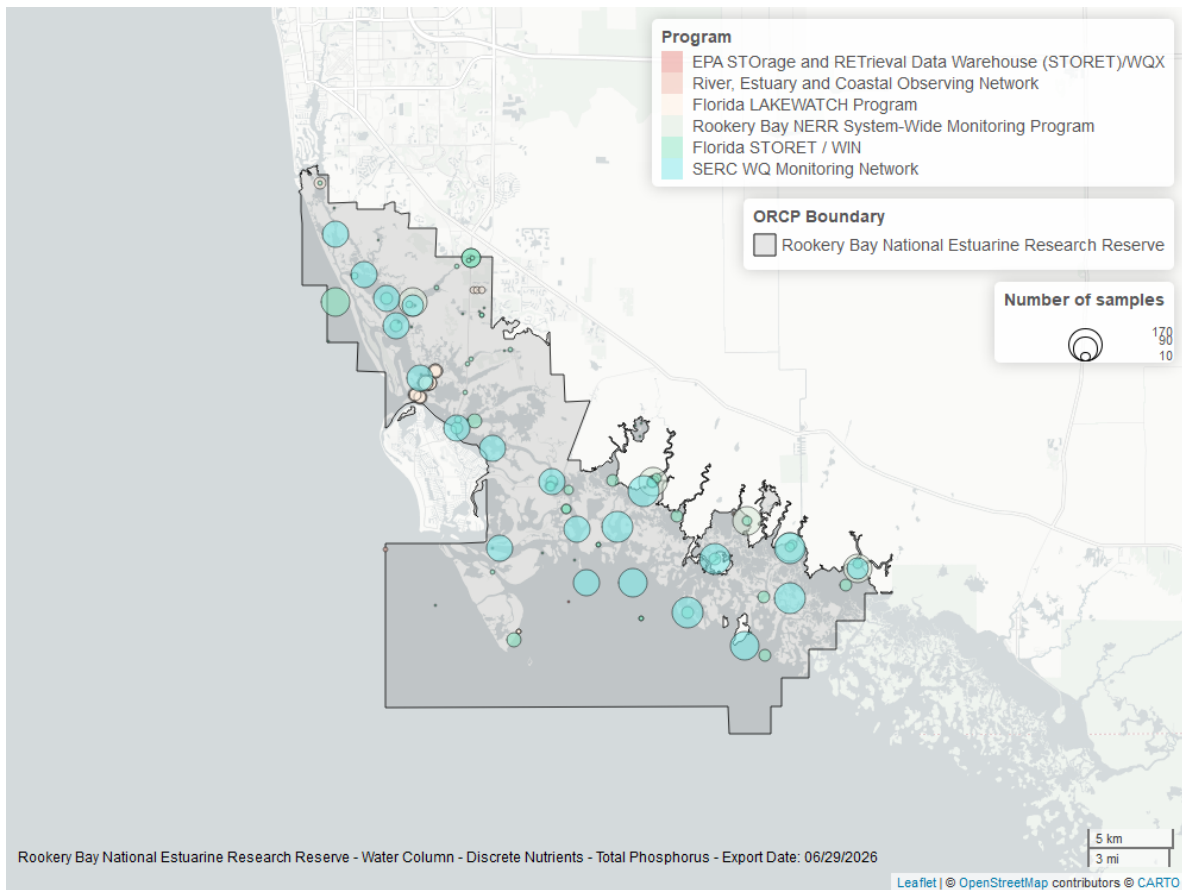


Figure 4: Map showing location of discrete water quality sampling locations within the boundaries of *Rookery Bay National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Quality

Dissolved Oxygen - Continuous

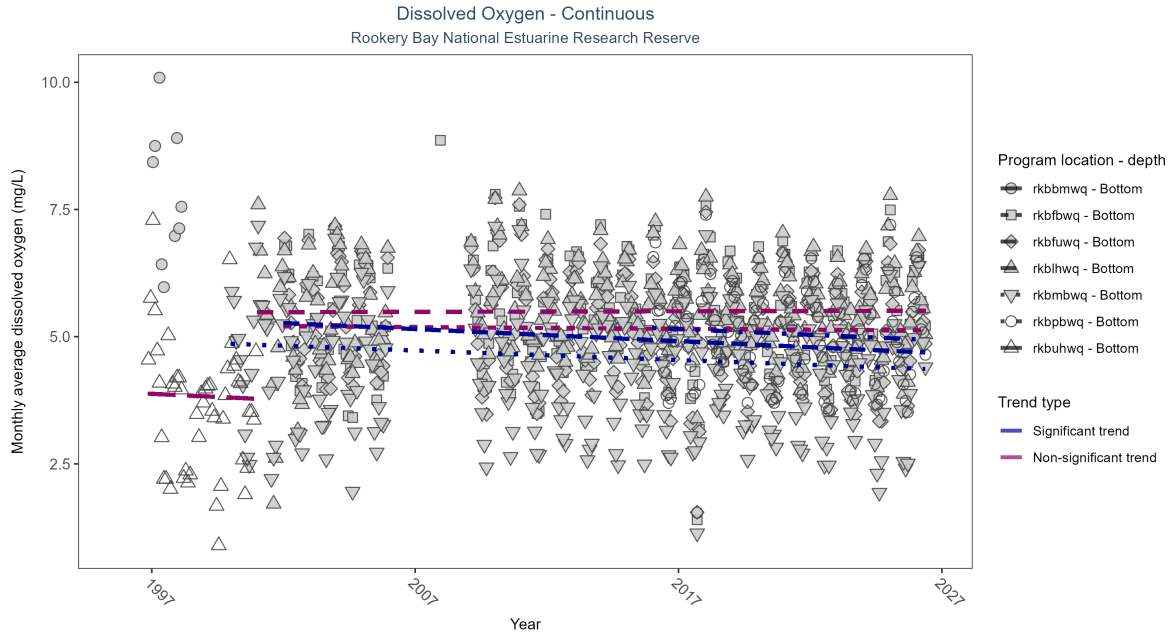


Figure 5: Scatter plot of monthly average dissolved oxygen over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 3: At three program locations, monthly average dissolved oxygen decreased between 0.02 and 0.02 mg/L per year. No detectable change in monthly average dissolved oxygen was observed at three locations. There was insufficient data to fit a model for one location.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
rkbbmwq	Insufficient data	10441	2	1997 - 1998	7.2	-	-	-	-
rkbfwq	No detectable trend	612445	23	2002 - 2026	5.4	-0.03	5.21	0	0.51
rkbfuwq	Decreasing trend	644958	22	2002 - 2026	5.1	-0.3	5.27	-0.02	0
rkblhwq	No detectable trend	622816	23	2001 - 2026	5.6	0.01	5.48	0	0.74
rkmbmwq	Decreasing trend	665650	24	2000 - 2026	4.4	-0.24	4.86	-0.02	0
rkbpbwq	Decreasing trend	341310	11	2016 - 2026	5.0	-0.15	5.17	-0.02	0.04
rkbuhwq	No detectable trend	58164	5	1996 - 2000	3.6	0.01	3.9	-0.02	1

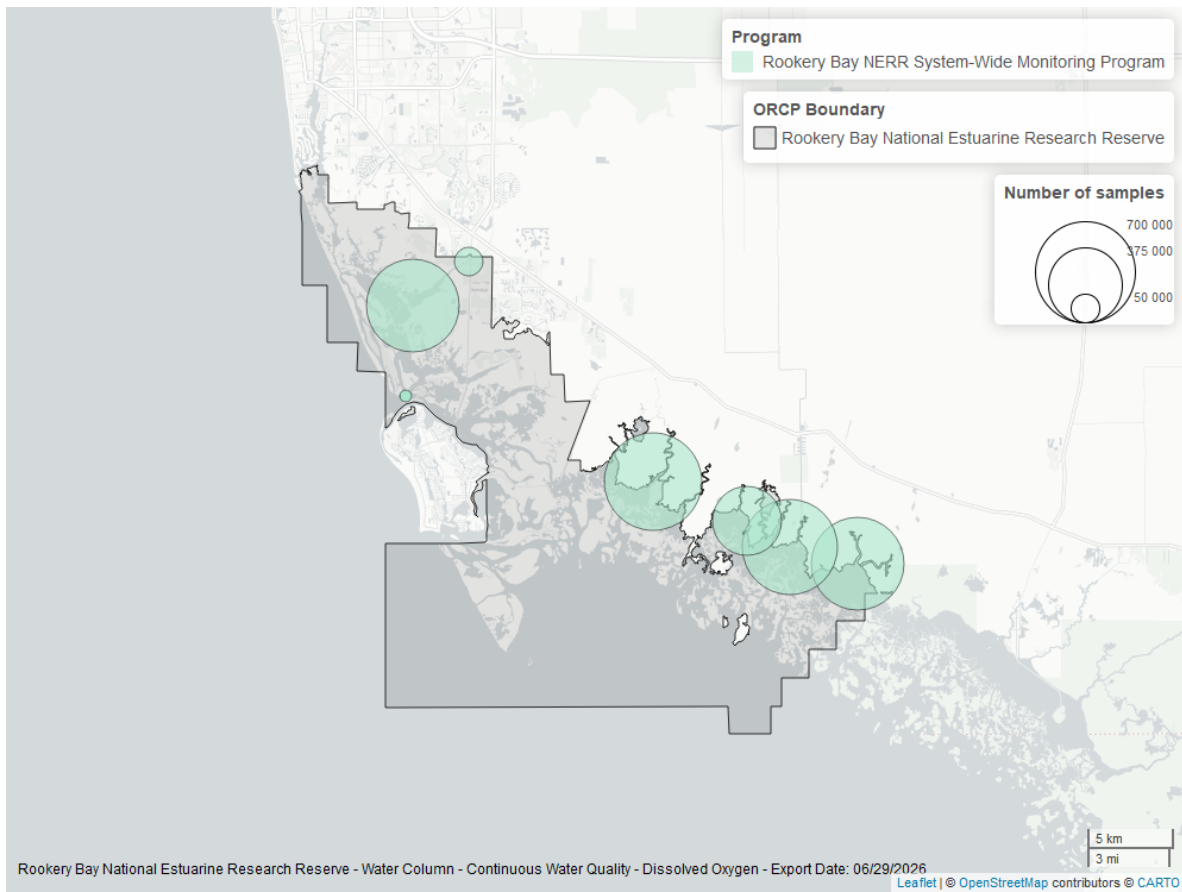


Figure 6: Map showing location of dissolved oxygen continuous water quality sampling locations within the boundaries of *Rookery Bay National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen - Discrete

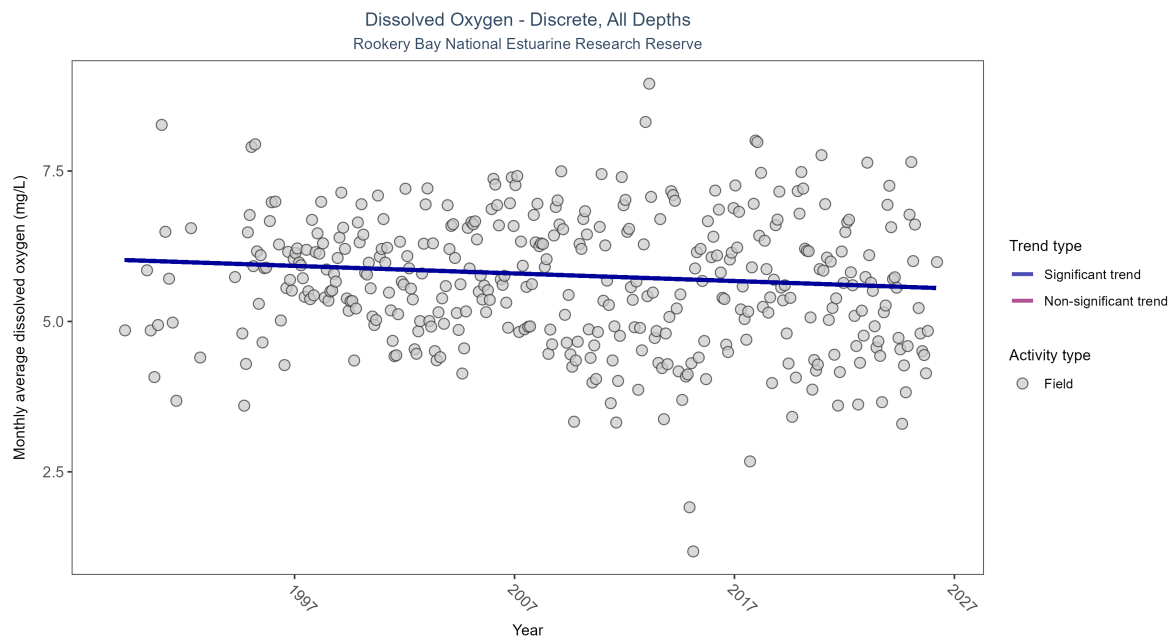


Figure 7: Scatter plot of monthly average dissolved oxygen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen values measured in the field (circles) are included in the plot.

Table 4: Monthly average dissolved oxygen decreased by 0.01 mg/L per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Decreasing trend	14827	37	1989 - 2026	5.8	-0.10648	6.02409	-0.01259	0.0029

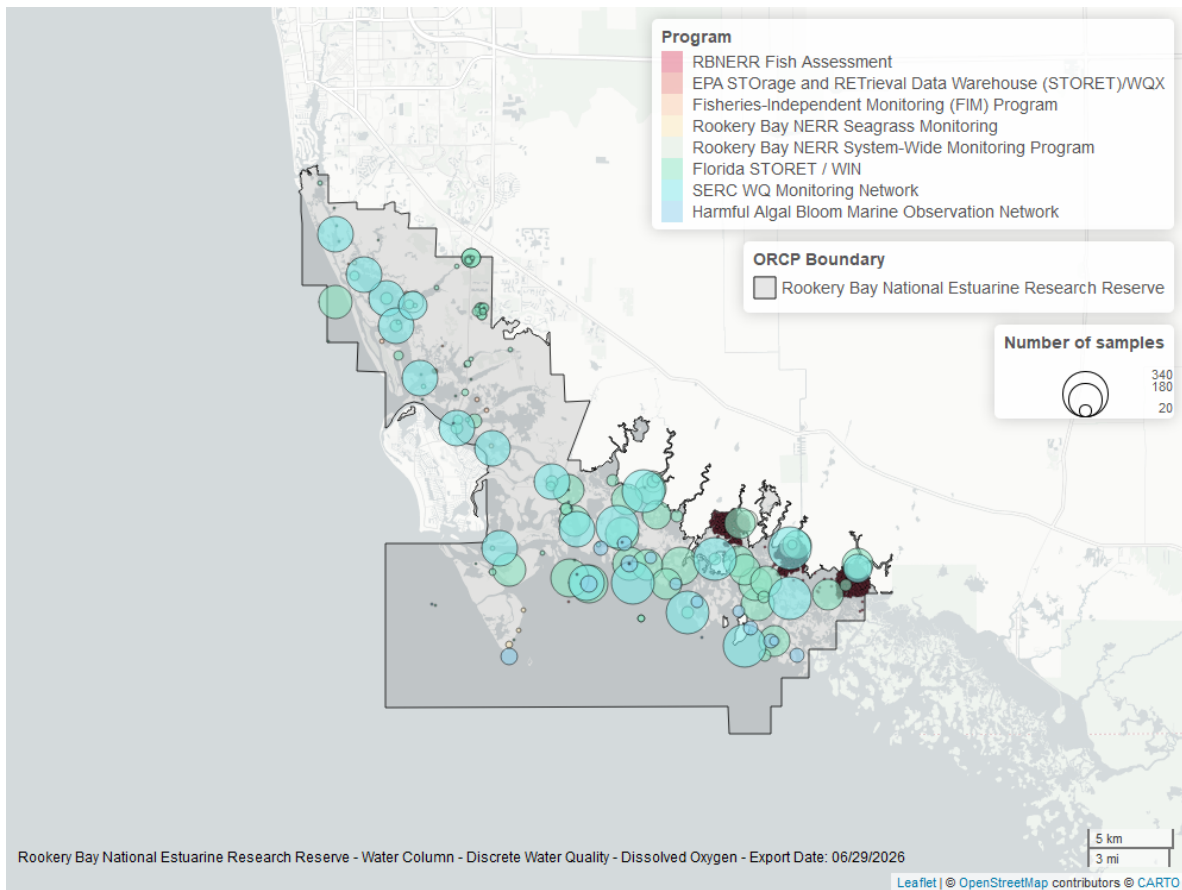


Figure 8: Map showing location of discrete water quality sampling locations within the boundaries of *Rookery Bay National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen Saturation - Continuous

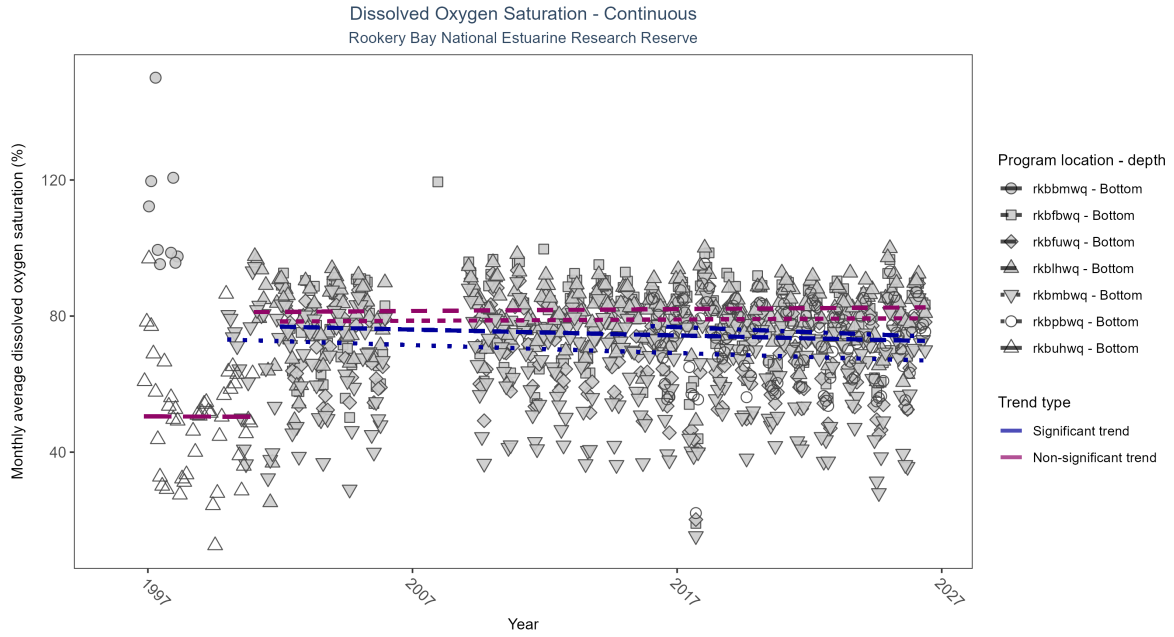


Figure 9: Scatter plot of monthly average dissolved oxygen saturation over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

\begin{table}[H] \caption{At three program locations, monthly average dissolved oxygen saturation decreased between 0.17 and 0.29% per year. No detectable change in monthly average dissolved oxygen saturation was observed at three locations. There was insufficient data to fit a model for one location.}

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
rkbbmwq	Insufficient data	10441	2	1997 - 1998	102.4	-	-	-	-
rkbfwq	No detectable trend	616751	23	2002 - 2026	79.1	0.03	78.42	0.04	0.51
rkbfuwq	Decreasing trend	645300	22	2002 - 2026	72.7	-0.2	76.89	-0.17	0
rkblhwq	No detectable trend	635134	23	2001 - 2026	82.1	0.05	81.17	0.05	0.22
rkblmwq	Decreasing trend	671974	24	2000 - 2026	65.4	-0.21	73.09	-0.23	0
rkbpbwq	Decreasing trend	344747	11	2016 - 2026	72.9	-0.15	77.04	-0.29	0.04
rkbuwq	No detectable trend	58164	5	1996 - 2000	49.7	0	50.53	-0.02	1

\end{table}

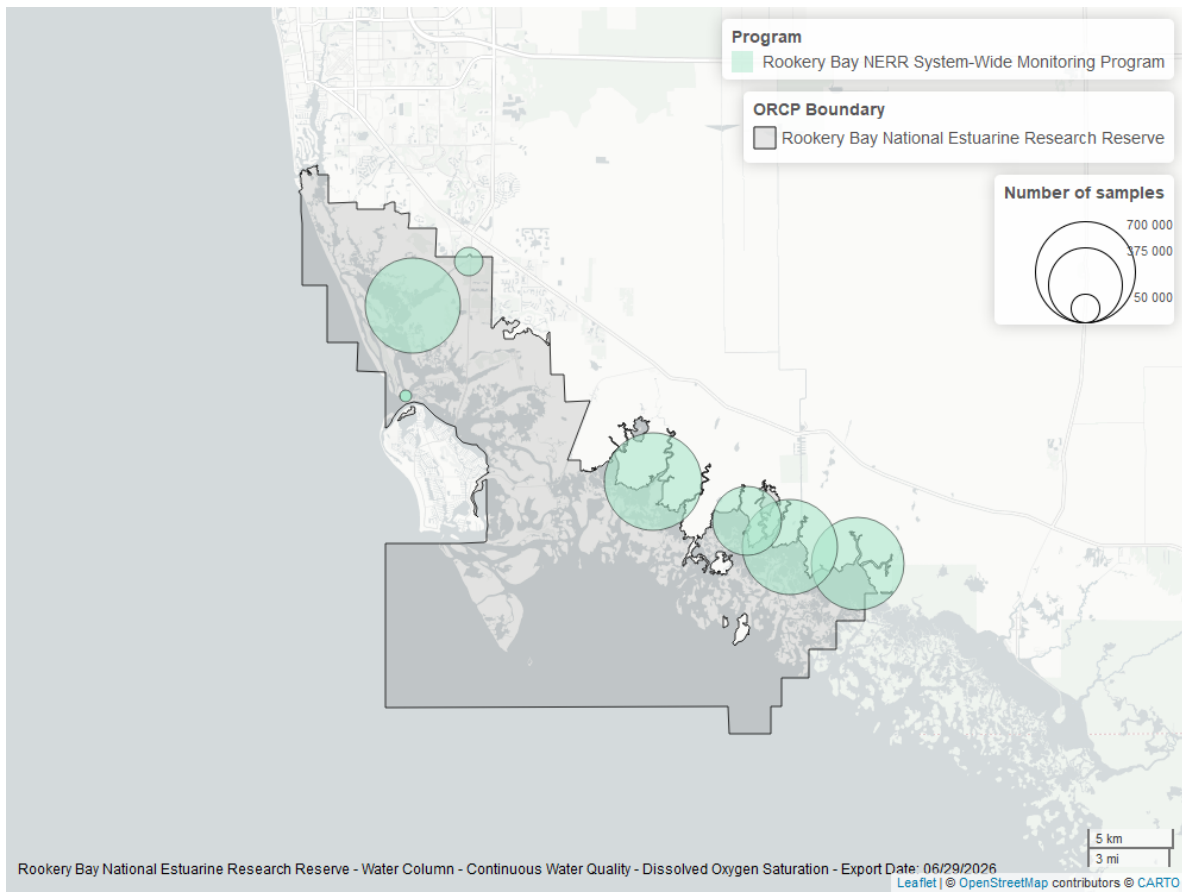


Figure 10: Map showing location of dissolved oxygen saturation continuous water quality sampling locations within the boundaries of *Rookery Bay National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen Saturation - Discrete

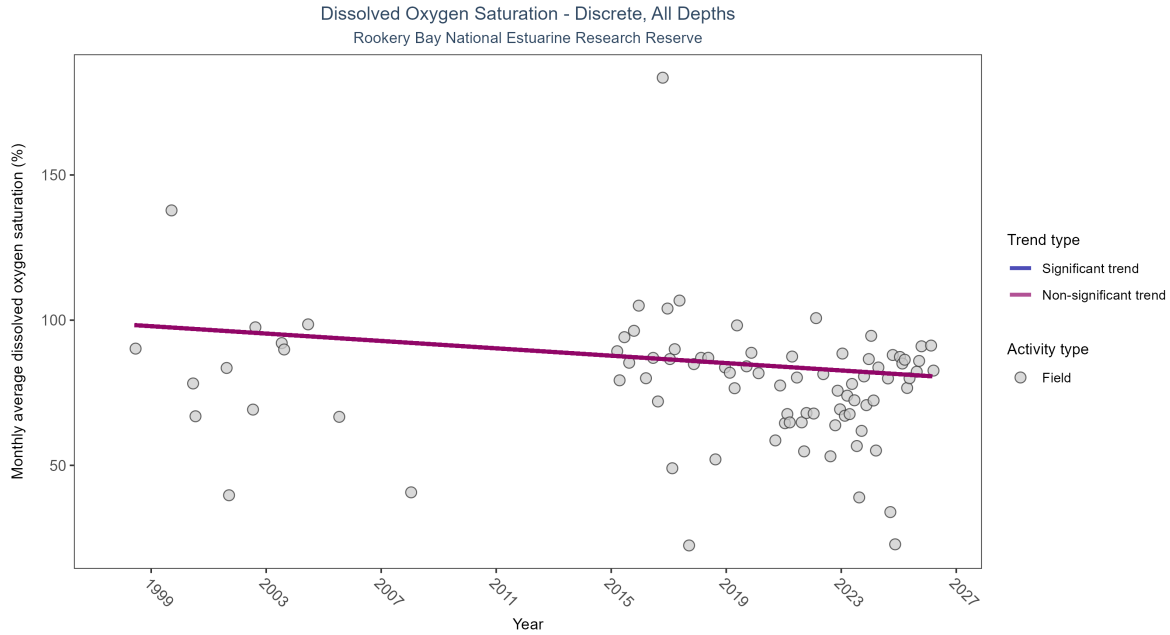


Figure 11: Scatter plot of monthly average dissolved oxygen saturation over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen saturation values measured in the field (circles) are included in the plot.

Table 5: Dissolved oxygen saturation showed no detectable trend between 1998 and 2026.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	No detectable trend	359	21	1998 - 2026	80.1	-0.21051	98.55709	-0.63528	0.0537

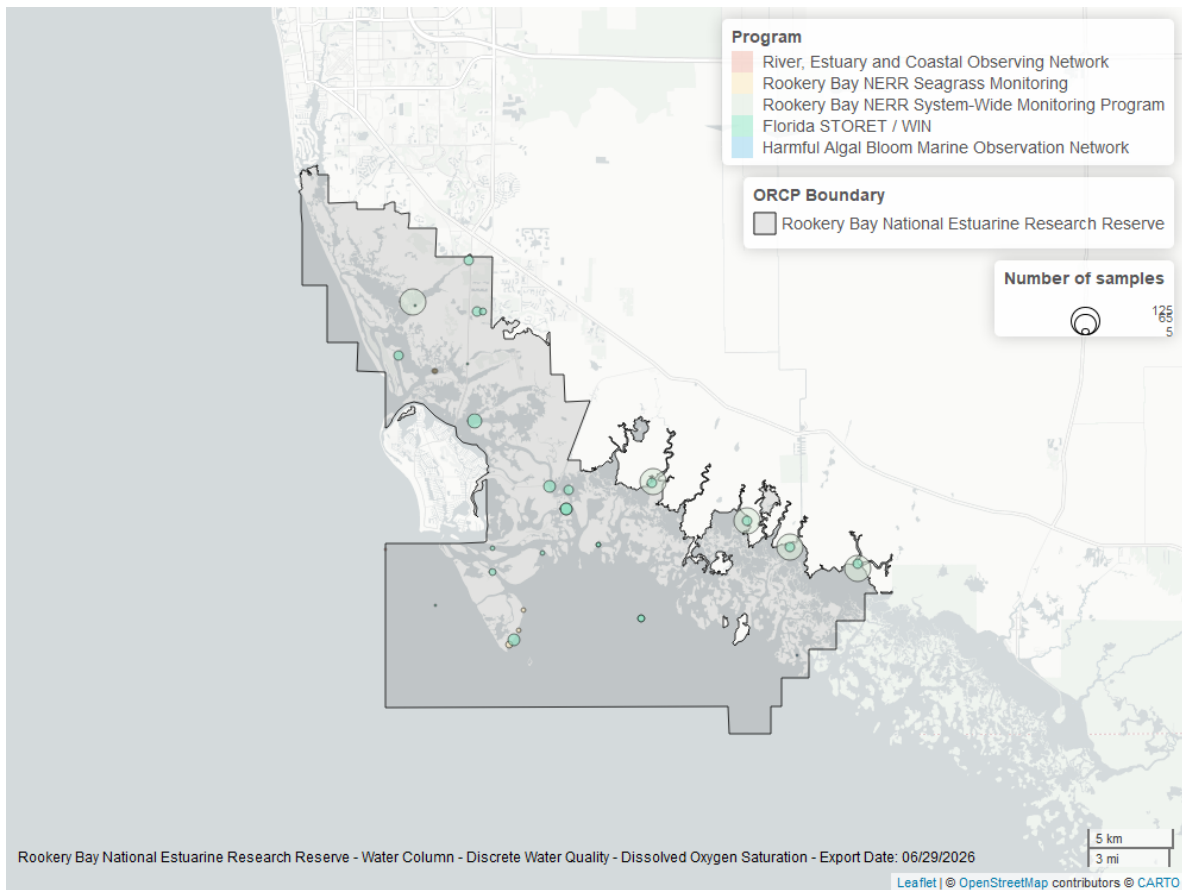


Figure 12: Map showing location of discrete water quality sampling locations within the boundaries of *Rookery Bay National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Salinity - Discrete

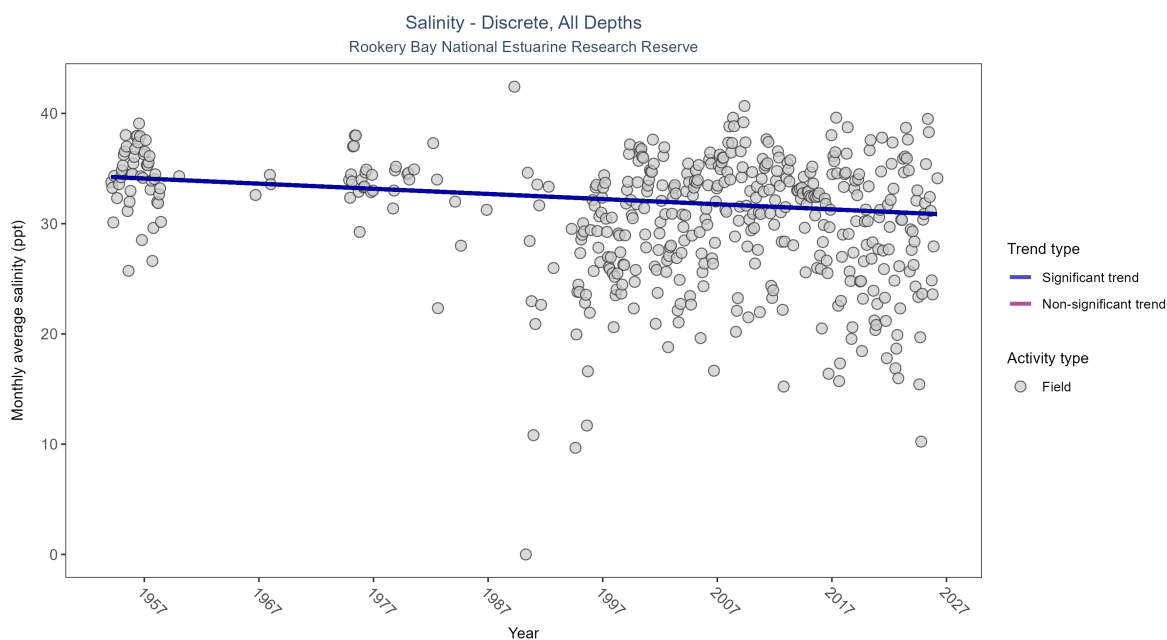


Figure 13: Scatter plot of monthly average salinity over time. If the time series included ten or more years of discrete observations, significant (blue) or non-significant (magenta) trend lines are also shown. Discrete salinity values derived from grab samples analyzed in the field (circles) or the laboratory (triangles) are both included in the plot.

Table 6: Monthly average salinity decreased by 0.05 ppt per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
All	Decreasing trend	16359	55	1954 - 2026	32.7	-0.16366	34.23535	-0.04666	0

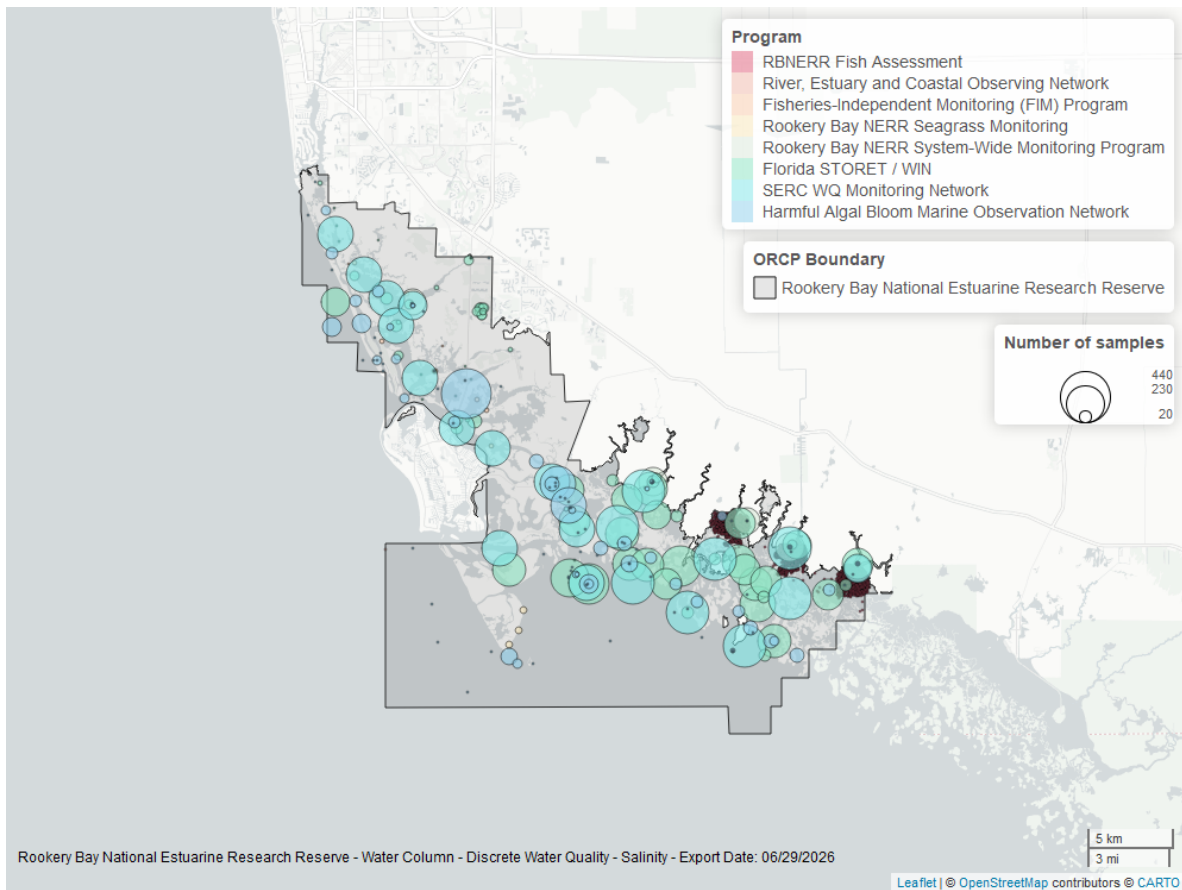


Figure 14: Map showing location of discrete water quality sampling locations within the boundaries of *Rookery Bay National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Salinity - Continuous

National Water Information System - 7

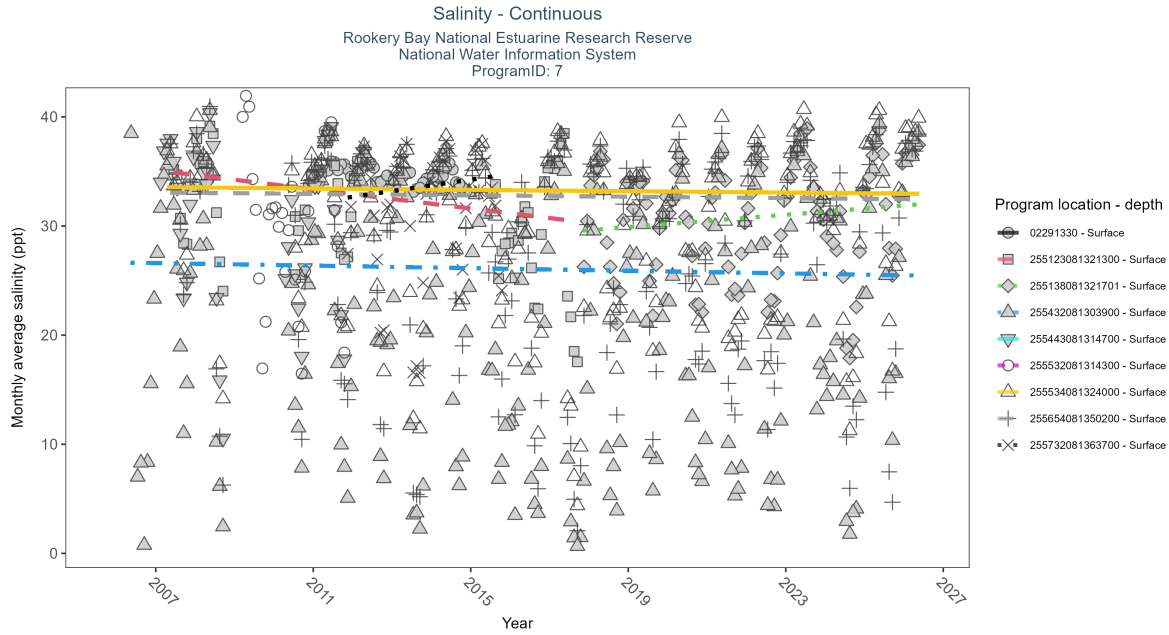


Figure 15: Scatter plot of monthly average salinity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 7: At three program locations, monthly average salinity increased between 0.25 and 2.06 ppt per year. At three program locations, monthly average salinity decreased between 0.06 and 0.44 ppt per year. No detectable change in monthly average salinity was observed at six locations. There was insufficient data to fit a model for four locations.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
02291330	Insufficient data	1697	4	2011 - 2014	35	-	-	-	-
255123081321300	Decreasing trend	1809	8	2007 - 2017	32	-0.23	35.11	-0.44	0.02
255138081321701	Increasing trend	3074	10	2017 - 2026	32	0.21	29.33	0.28	0.01
255432081303900	No detectable trend	6556	20	2006 - 2026	22	-0.05	26.67	-0.06	0.29
255443081314700	Insufficient data	1465	4	2007 - 2011	32	-	-	-	-
255532081314300	Insufficient data	902	3	2009 - 2011	31	-	-	-	-
255534081324000	No detectable trend	6250	19	2007 - 2026	32	-0.02	33.56	-0.03	0.62
255654081350200	No detectable trend	6271	19	2007 - 2026	32	-0.03	33.07	-0.03	0.52
255732081363700	No detectable trend	1434	5	2011 - 2015	34	0.25	32.15	0.52	0.1

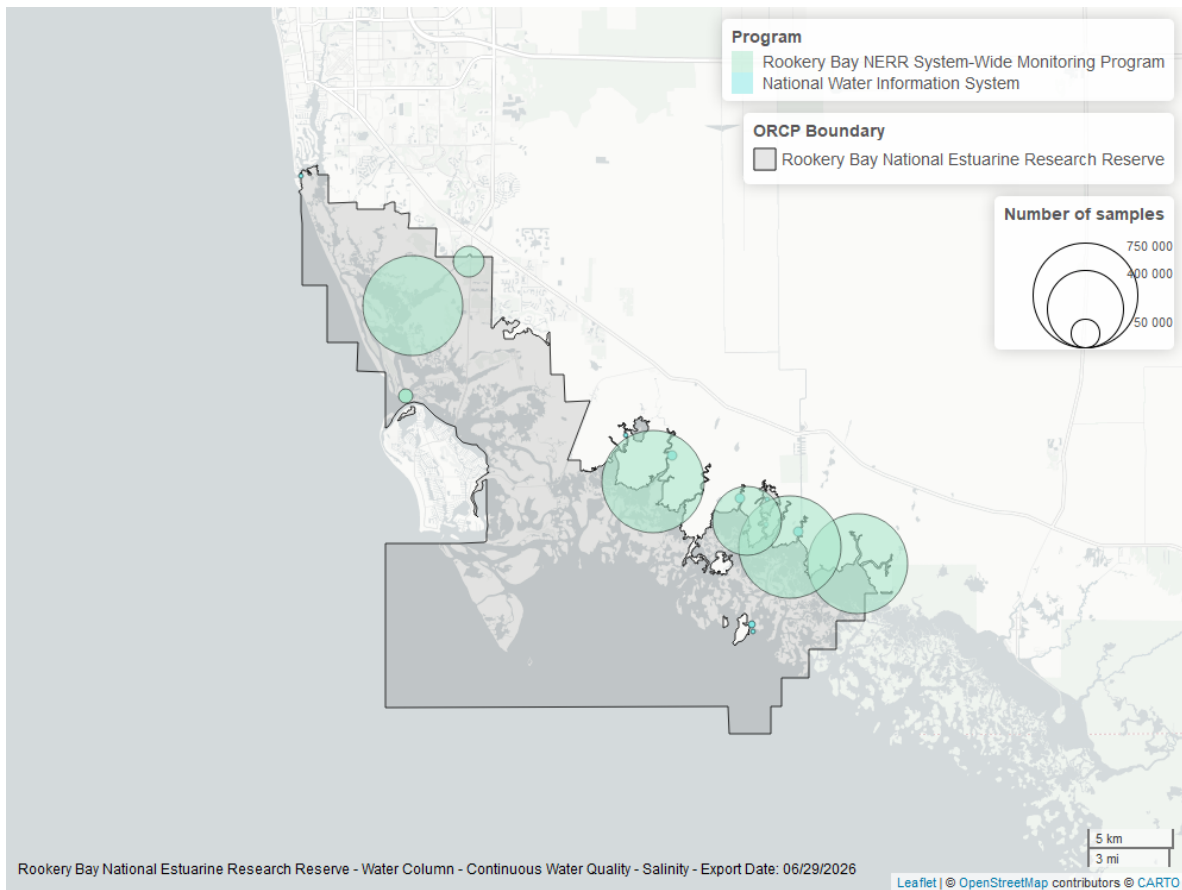


Figure 16: Map showing location of salinity continuous water quality sampling locations within the boundaries of *Rookery Bay National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

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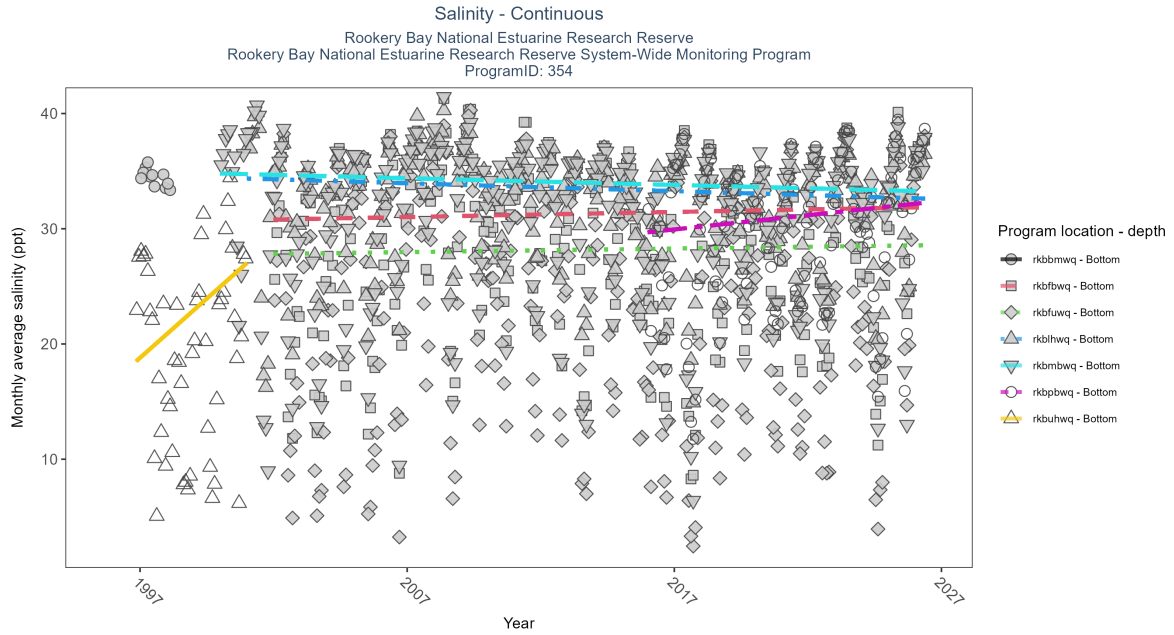


Figure 17: Scatter plot of monthly average salinity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 8: At three program locations, monthly average salinity increased between 0.25 and 2.06 ppt per year. At three program locations, monthly average salinity decreased between 0.06 and 0.44 ppt per year. No detectable change in monthly average salinity was observed at six locations. There was insufficient data to fit a model for four locations.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
rkbbmwq	Insufficient data	12256	2	1997 - 1998	34.5	-	-	-	-
rkbfwq	No detectable trend	711718	25	2002 - 2026	30.0	0.05	30.8	0.04	0.21
rkbfuwq	No detectable trend	725512	25	2002 - 2026	26.5	0.03	27.82	0.03	0.51
rkblhwq	Decreasing trend	709967	26	2001 - 2026	33.2	-0.12	34.36	-0.07	0
rkbmbwq	Decreasing trend	744660	27	2000 - 2026	33.4	-0.09	34.8	-0.06	0.02
rkbpbwq	Increasing trend	341625	11	2016 - 2026	30.7	0.17	29.69	0.25	0.02
rkbulhwq	Increasing trend	68446	5	1996 - 2000	21.1	0.31	16.74	2.06	0.04

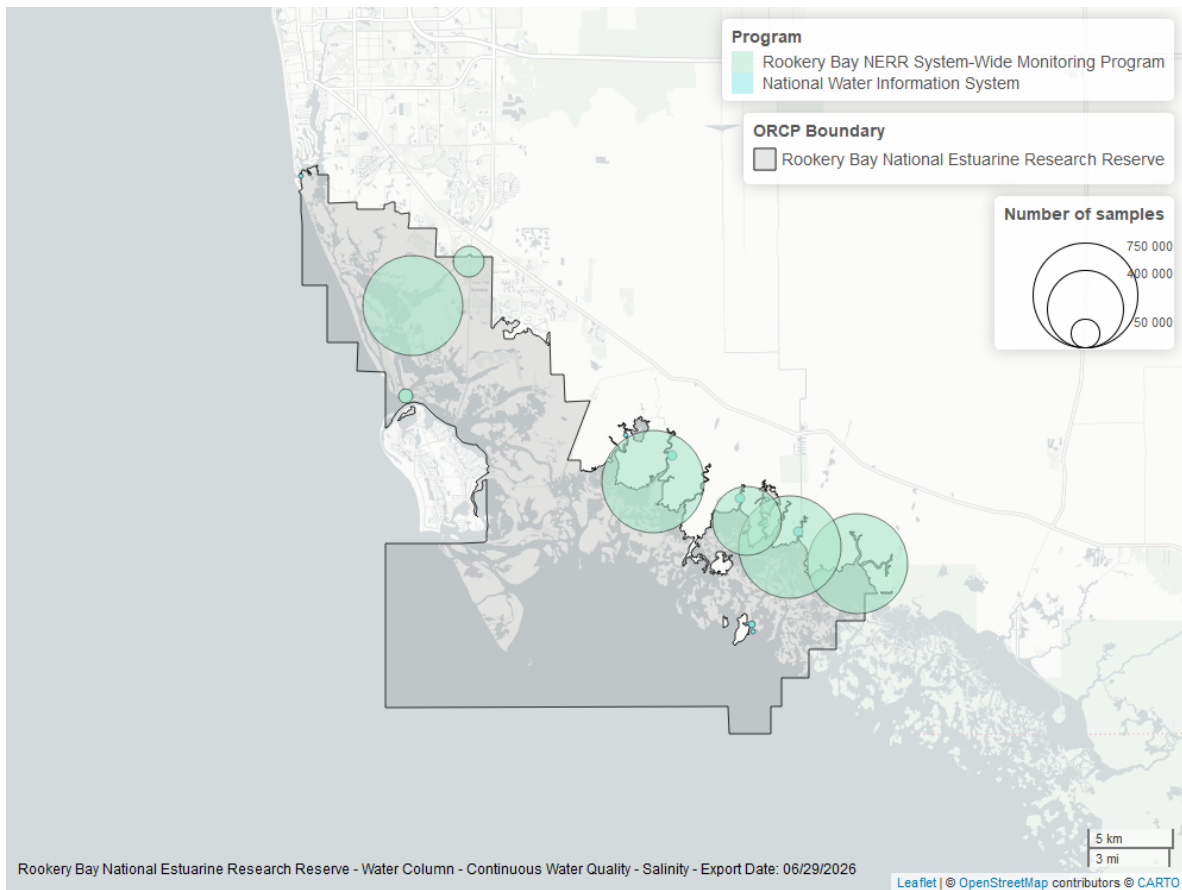


Figure 18: Map showing location of salinity continuous water quality sampling locations within the boundaries of *Rookery Bay National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Temperature - Continuous National Water Information System - 7

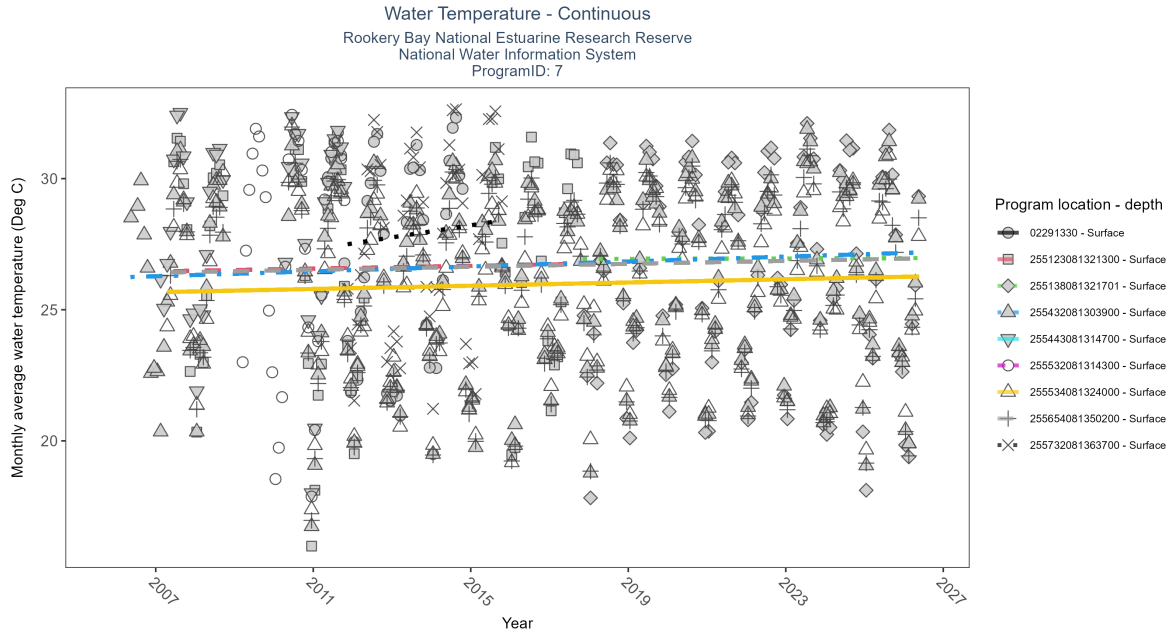


Figure 19: Scatter plot of monthly average water temperature over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 9: At seven program locations, monthly average water temperature increased between 0.03 and 0.06Deg C per year. No detectable change in monthly average water temperature was observed at five locations. There was insufficient data to fit a model for four locations.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
02291330	Insufficient data	1901	4	2011 - 2014	28.3	-	-	-	-
255123081321300	No detectable trend	1818	8	2007 - 2017	27.3	0.1	26.46	0.03	0.38
255138081321701	No detectable trend	3093	10	2017 - 2026	27.0	0.01	26.94	0	0.97
255432081303900	Increasing trend	6615	20	2006 - 2026	27.0	0.17	26.23	0.05	0
255443081314700	Insufficient data	2011	4	2007 - 2011	29.3	-	-	-	-
255532081314300	Insufficient data	906	3	2009 - 2011	29.3	-	-	-	-
255534081324000	Increasing trend	6313	19	2007 - 2026	26.6	0.13	25.67	0.03	0.01
255654081350200	Increasing trend	6310	19	2007 - 2026	26.7	0.11	26.41	0.03	0.03
255732081363700	No detectable trend	1435	5	2011 - 2015	28.4	0.24	27.3	0.23	0.11

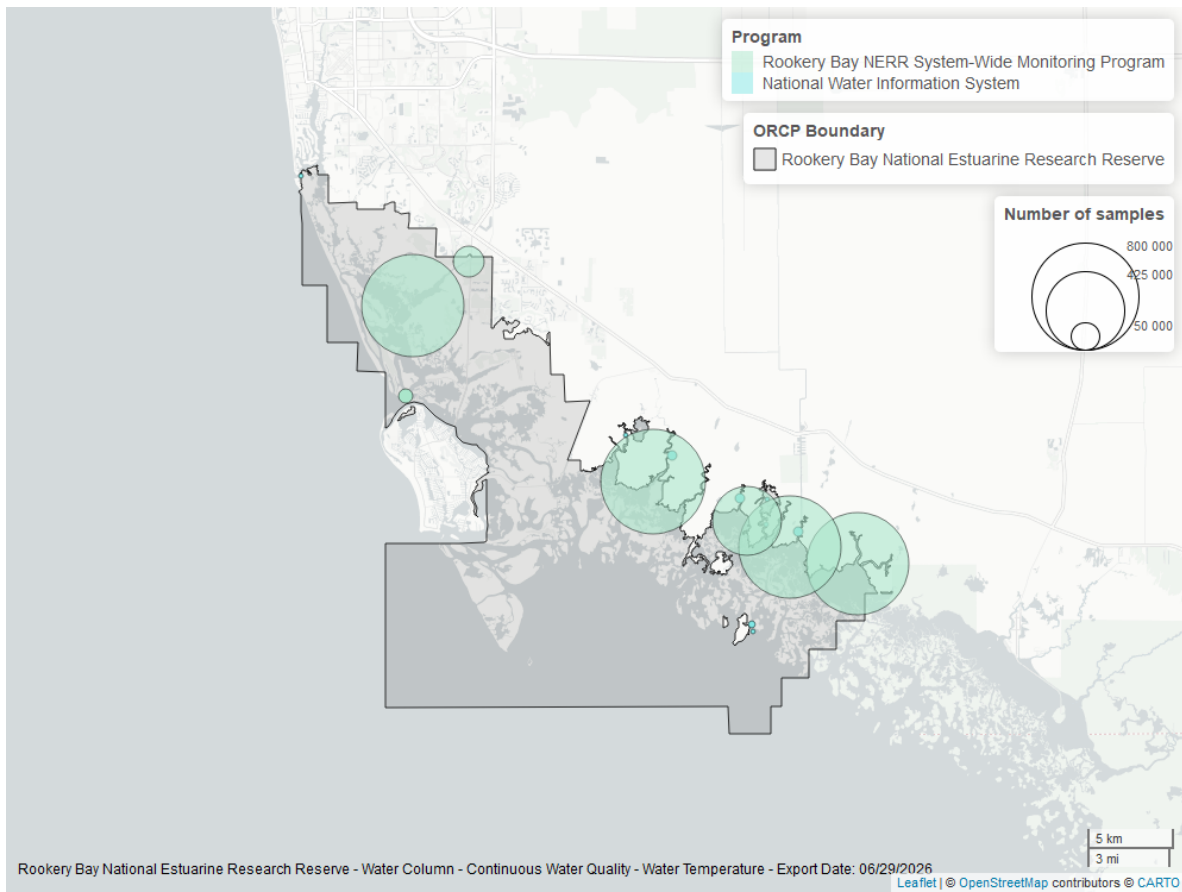


Figure 20: Map showing location of water temperature continuous water quality sampling locations within the boundaries of *Rookery Bay National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

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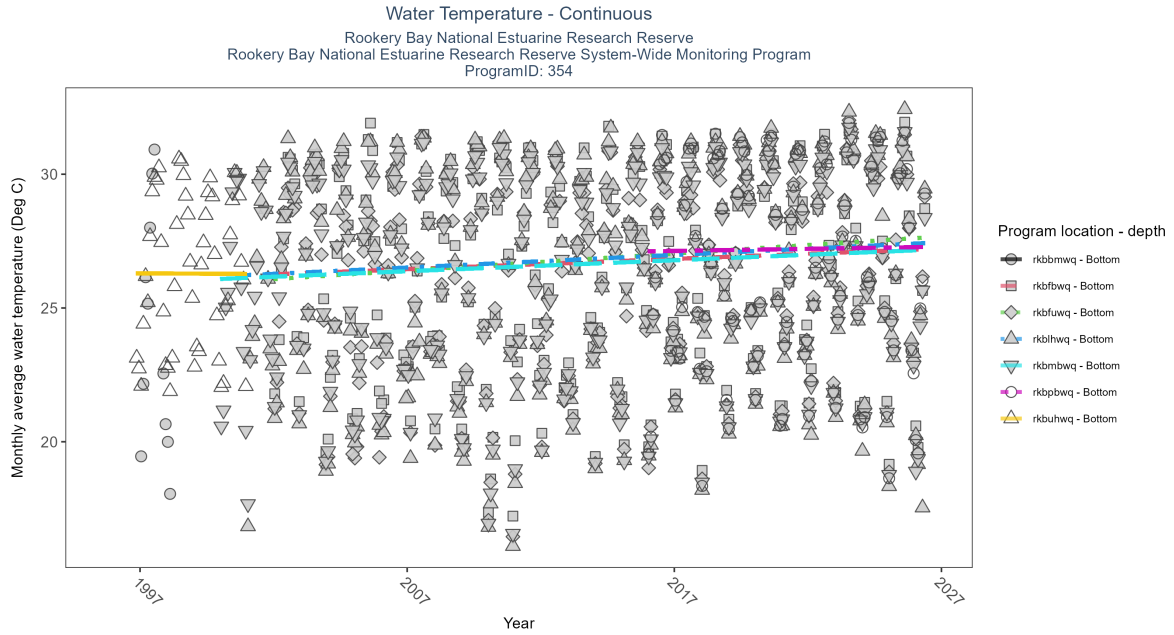


Figure 21: Scatter plot of monthly average water temperature over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 10: At seven program locations, monthly average water temperature increased between 0.03 and 0.06Deg C per year. No detectable change in monthly average water temperature was observed at five locations. There was insufficient data to fit a model for four locations.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
rkbbmwq	Insufficient data	12610	2	1997 - 1998	23.8	-	-	-	-
rkbbfwq	Increasing trend	727557	25	2002 - 2026	27.0	0.22	26.26	0.04	0
rkbfuwq	Increasing trend	736040	25	2002 - 2026	26.9	0.26	26.08	0.06	0
rkblhwq	Increasing trend	741119	26	2001 - 2026	26.8	0.23	26.23	0.05	0
rkmbbwq	Increasing trend	770564	27	2000 - 2026	26.8	0.24	26.09	0.04	0
rkbpbwq	No detectable trend	346335	11	2016 - 2026	27.3	0.05	27.12	0.02	0.53
rkbuhwq	No detectable trend	68971	5	1996 - 2000	26.8	-0.01	26.3	-0.01	1

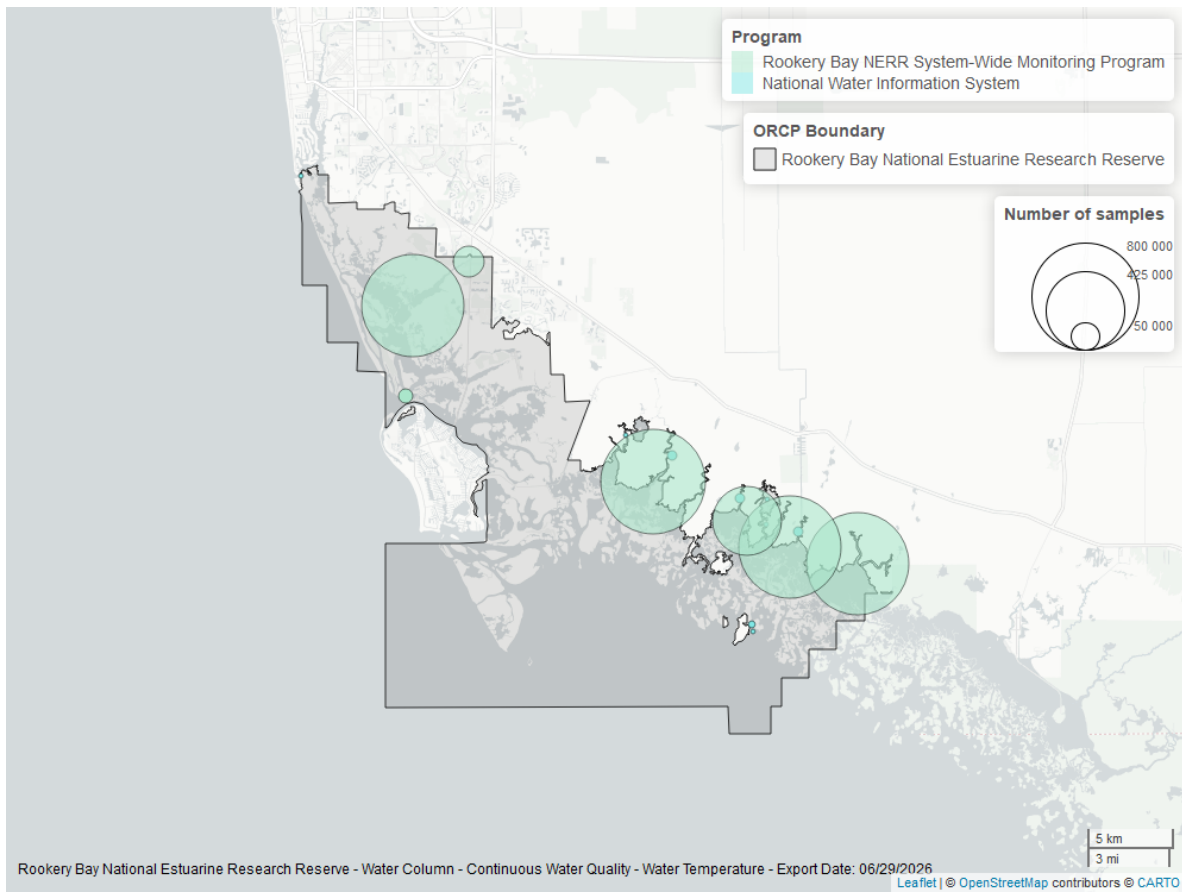


Figure 22: Map showing location of water temperature continuous water quality sampling locations within the boundaries of *Rookery Bay National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Temperature - Discrete

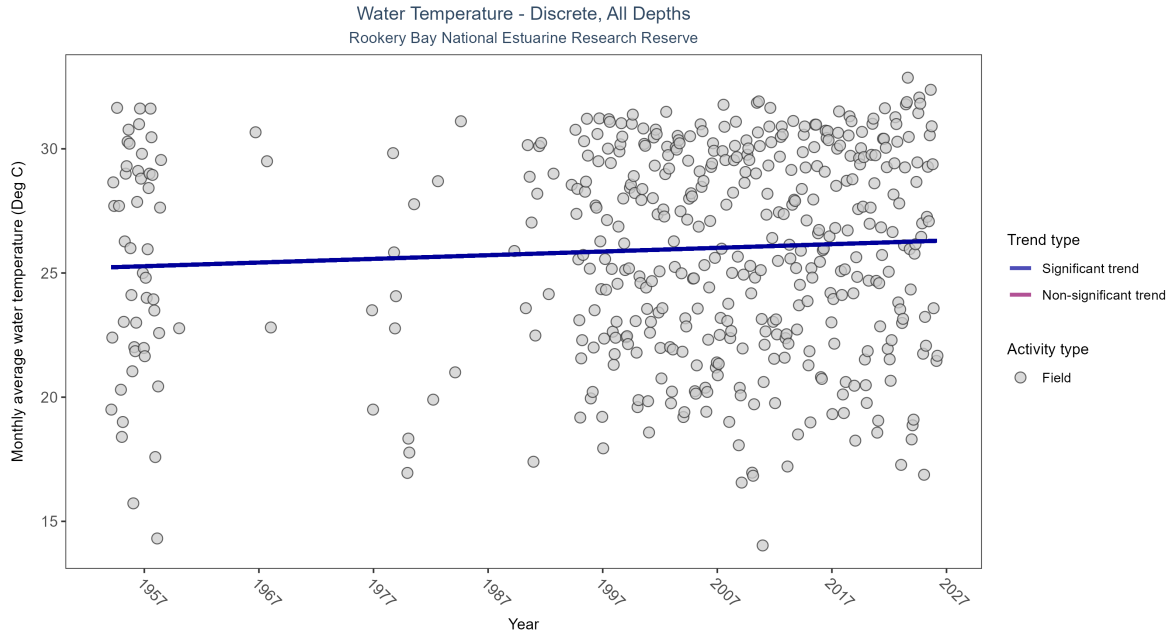


Figure 23: Scatter plot of monthly average water temperature over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only water temperature measurements taken in the field (circles) are included in the plot.

Table 11: Monthly average water temperature increased by 0.01Deg C per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Increasing trend	16380	52	1954 - 2026	26.645	0.1236	25.22869	0.0148	0.0001

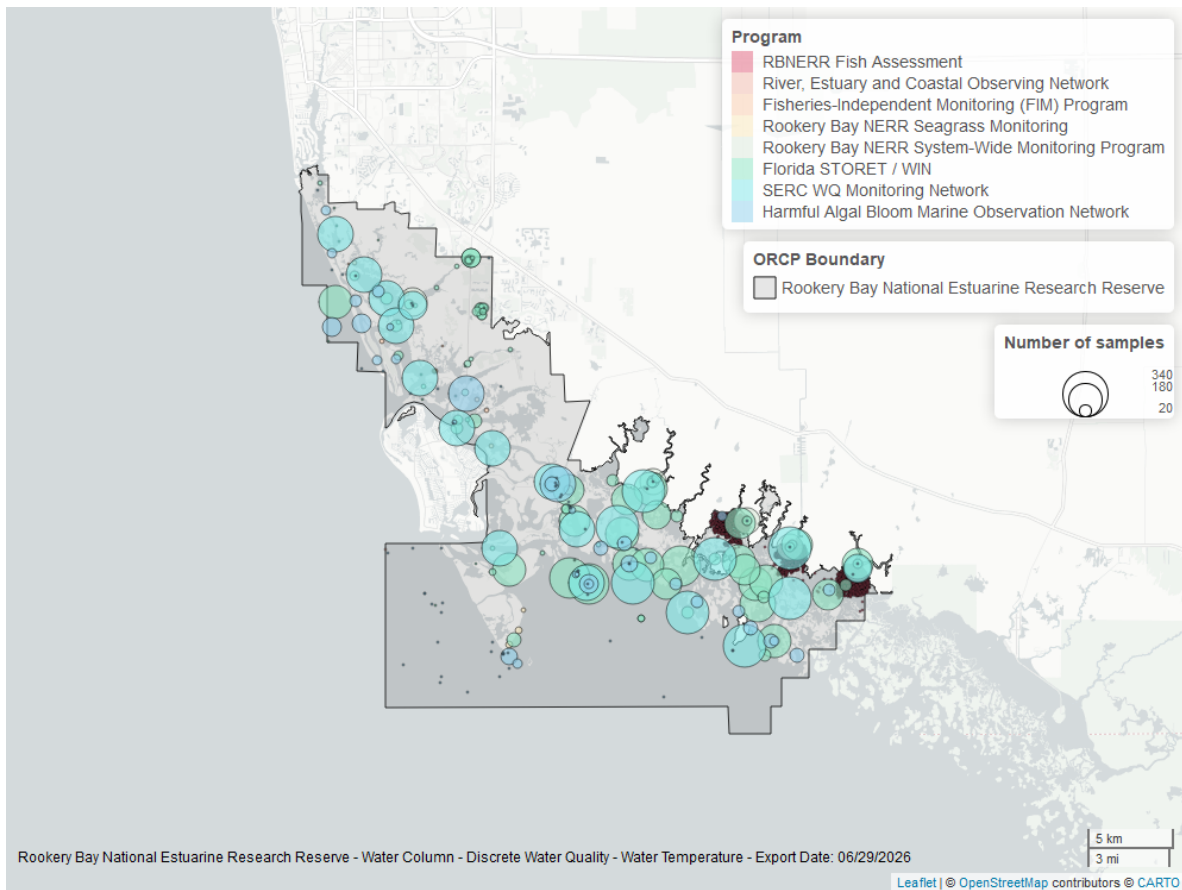


Figure 24: Map showing location of discrete water quality sampling locations within the boundaries of *Rookery Bay National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

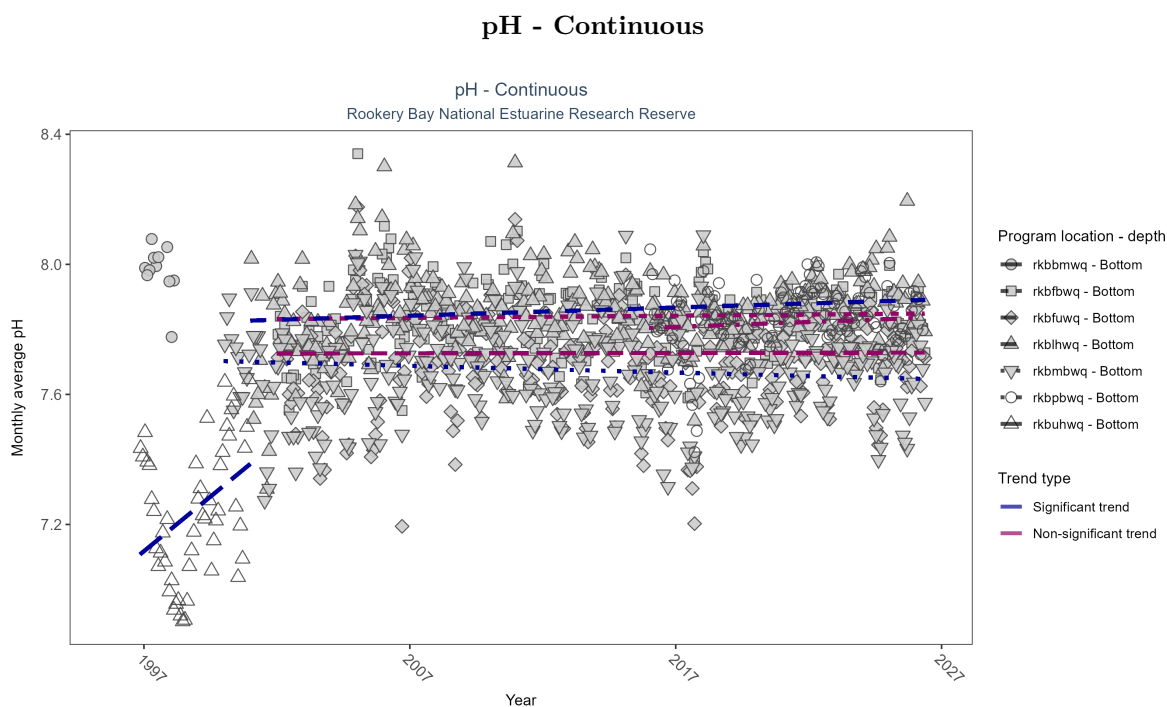


Figure 25: Scatter plot of monthly average pH over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 12: At two program locations, monthly average pH increased by less than 0.01 pH units per year at one site and by 0.07 pH units per year at the other. At one program location, monthly average pH decreased by less than 0.01 pH units per year. No detectable change in monthly average pH was observed at three locations. There was insufficient data to fit a model for one location.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
rkbmbwq	Insufficient data	12610	2	1997 - 1998	8.0	-	-	-	-
rkbfbwq	No detectable trend	679220	25	2002 - 2026	7.8	0.04	7.83	0	0.36
rkbflwq	No detectable trend	697654	25	2002 - 2026	7.7	0.01	7.73	0	0.88
rkbllwq	Increasing trend	680804	26	2001 - 2026	7.9	0.14	7.83	0	0
rkbmbwq	Decreasing trend	735914	27	2000 - 2026	7.7	-0.1	7.7	0	0.01
rkbpbwq	No detectable trend	332136	11	2016 - 2026	7.8	0.1	7.8	0	0.19
rkbuhwq	Increasing trend	65814	5	1996 - 2000	7.2	0.37	7.05	0.07	0.01

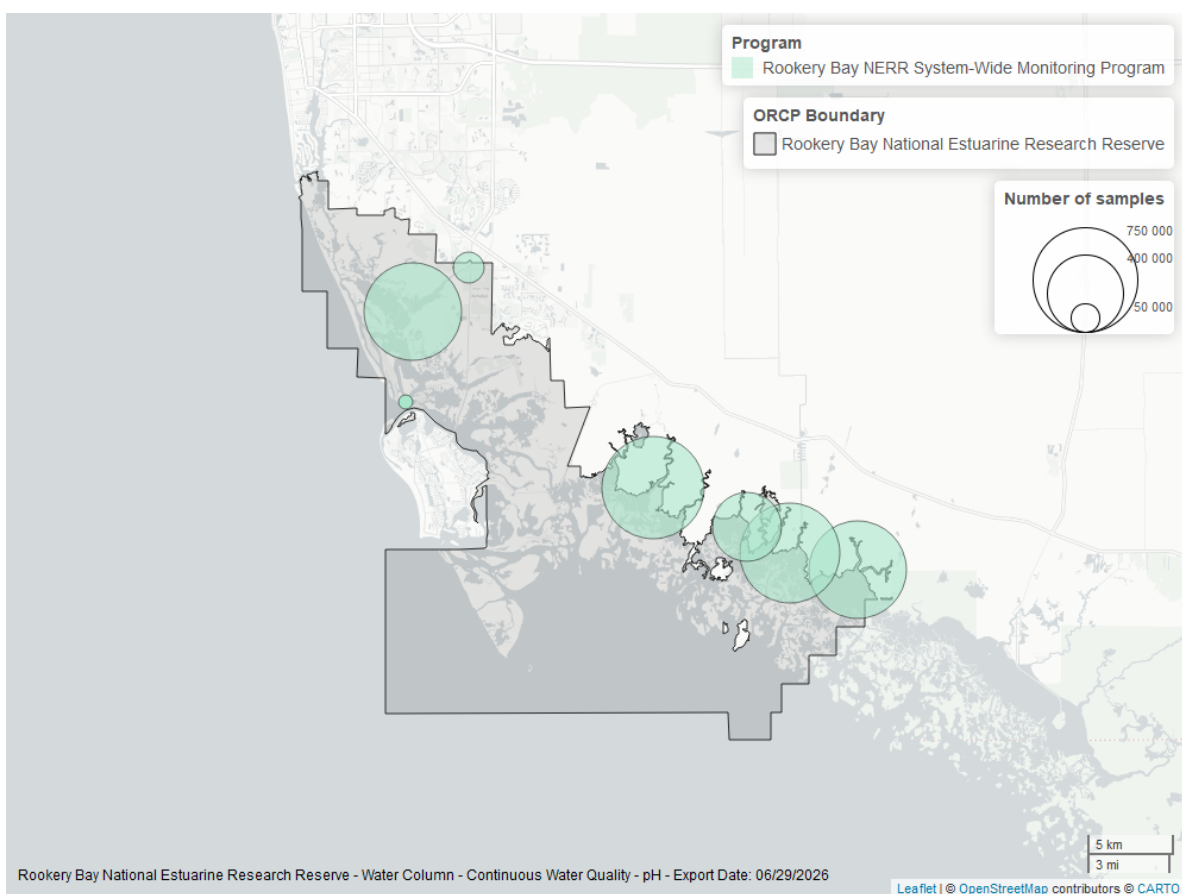


Figure 26: Map showing location of pH continuous water quality sampling locations within the boundaries of *Rookery Bay National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

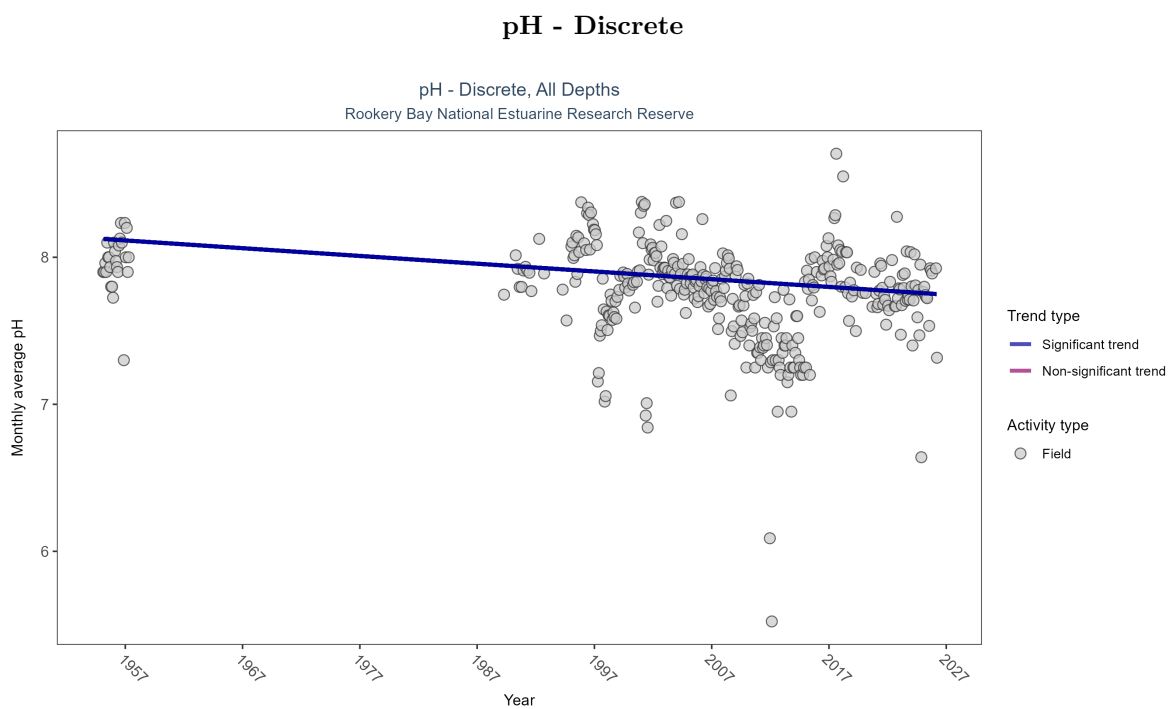


Figure 27: Scatter plot of monthly average pH over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only pH values measured in the field (circles) are included in the plot.

Table 13: Monthly average pH decreased by 0.01 pH units per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Decreasing trend	6499	40	1955 - 2026	7.88	-0.21498	8.12503	-0.00527	0

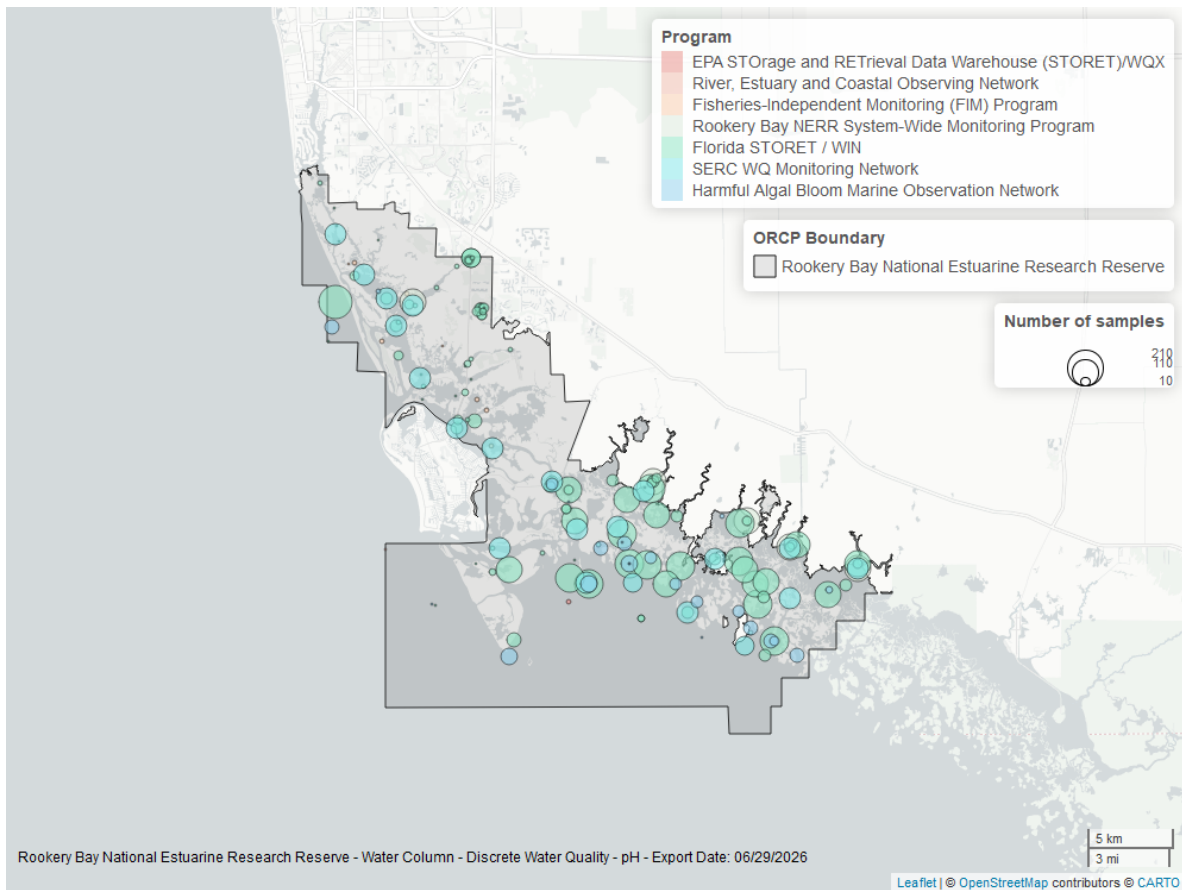


Figure 28: Map showing location of discrete water quality sampling locations within the boundaries of *Rookery Bay National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Specific Conductivity - Continuous

National Water Information System - 7

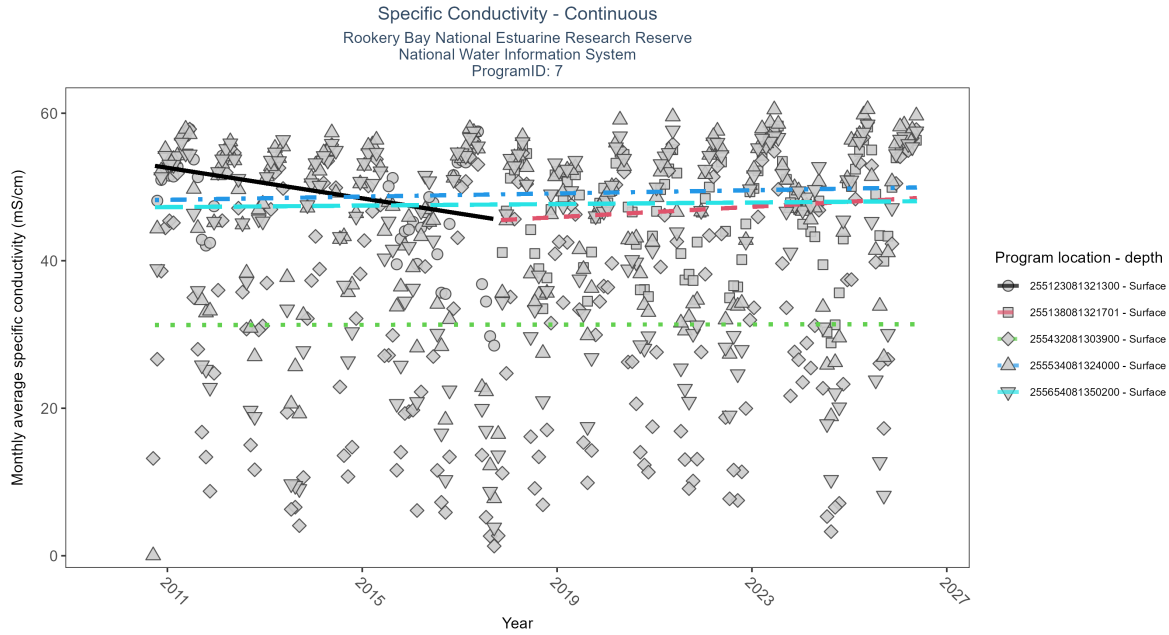


Figure 29: Scatter plot of monthly average specific conductivity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 14: At two program locations, monthly average specific conductivity increased by 0.32 mS/cm per year at one site and by 0.35 mS/cm per year at the other. At two program locations, monthly average specific conductivity decreased by 0.07 mS/cm per year at one site and by 0.09 mS/cm per year at the other. No detectable change in monthly average specific conductivity was observed at seven locations. There was insufficient data to fit a model for one location.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
255123081321300	No detectable trend	1295	6	2010 - 2017	49.1	-0.20	53.63	-1.03	0.20
255138081321701	Increasing trend	3074	10	2017 - 2026	48.7	0.21	45.21	0.35	0.01
255432081303900	No detectable trend	5600	17	2010 - 2026	34.9	0.00	31.28	0.01	0.98
255534081324000	No detectable trend	5597	17	2010 - 2026	48.8	0.06	48.14	0.11	0.24
255654081350200	No detectable trend	5635	17	2010 - 2026	48.8	0.04	47.21	0.05	0.43

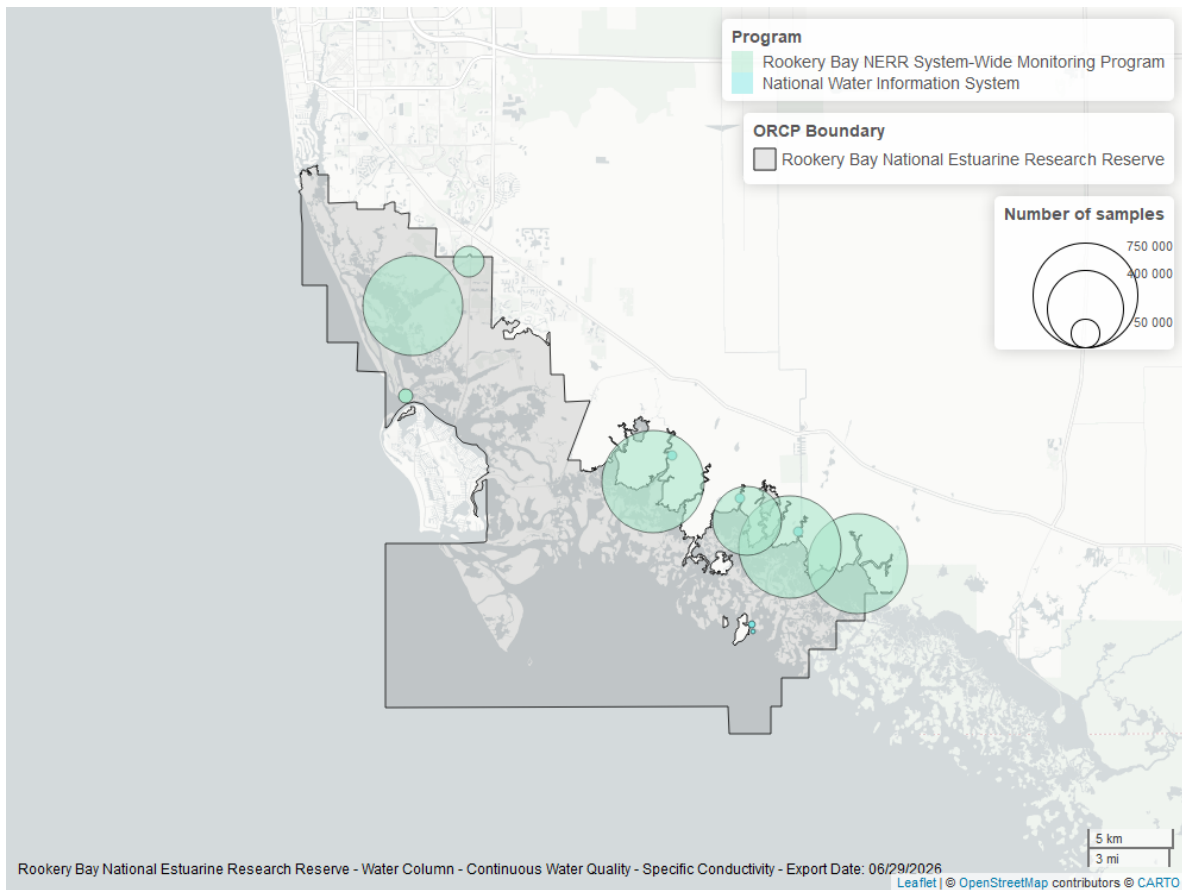


Figure 30: Map showing location of specific conductivity continuous water quality sampling locations within the boundaries of *Rookery Bay National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

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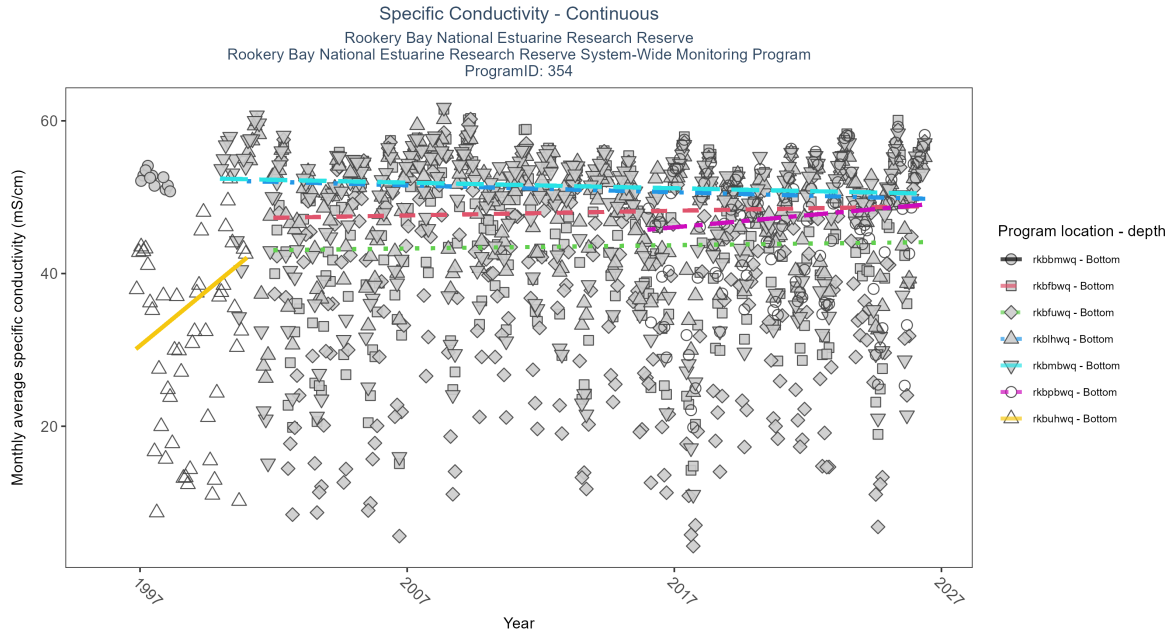


Figure 31: Scatter plot of monthly average specific conductivity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 15: At two program locations, monthly average specific conductivity increased by 0.32 mS/cm per year at one site and by 0.35 mS/cm per year at the other. At two program locations, monthly average specific conductivity decreased by 0.07 mS/cm per year at one site and by 0.09 mS/cm per year at the other. No detectable change in monthly average specific conductivity was observed at seven locations. There was insufficient data to fit a model for one location.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
rkbbmwq	Insufficient data	12256	2	1997 - 1998	52.41	-	-	-	-
rkbbfwq	No detectable trend	711718	25	2002 - 2026	46.24	0.05	47.3	0.06	0.2
rkbfuwq	No detectable trend	725511	25	2002 - 2026	41.45	0.03	43.06	0.04	0.51
rkblhwq	Decreasing trend	710006	26	2001 - 2026	50.68	-0.11	52.11	-0.09	0.01
rkmbmwq	Decreasing trend	744660	27	2000 - 2026	50.93	-0.09	52.43	-0.07	0.03
rkbpbwq	Increasing trend	341625	11	2016 - 2026	47.30	0.16	45.74	0.32	0.02
rkbuhwq	No detectable trend	68971	5	1996 - 2000	33.92	0.29	27.67	2.88	0.06

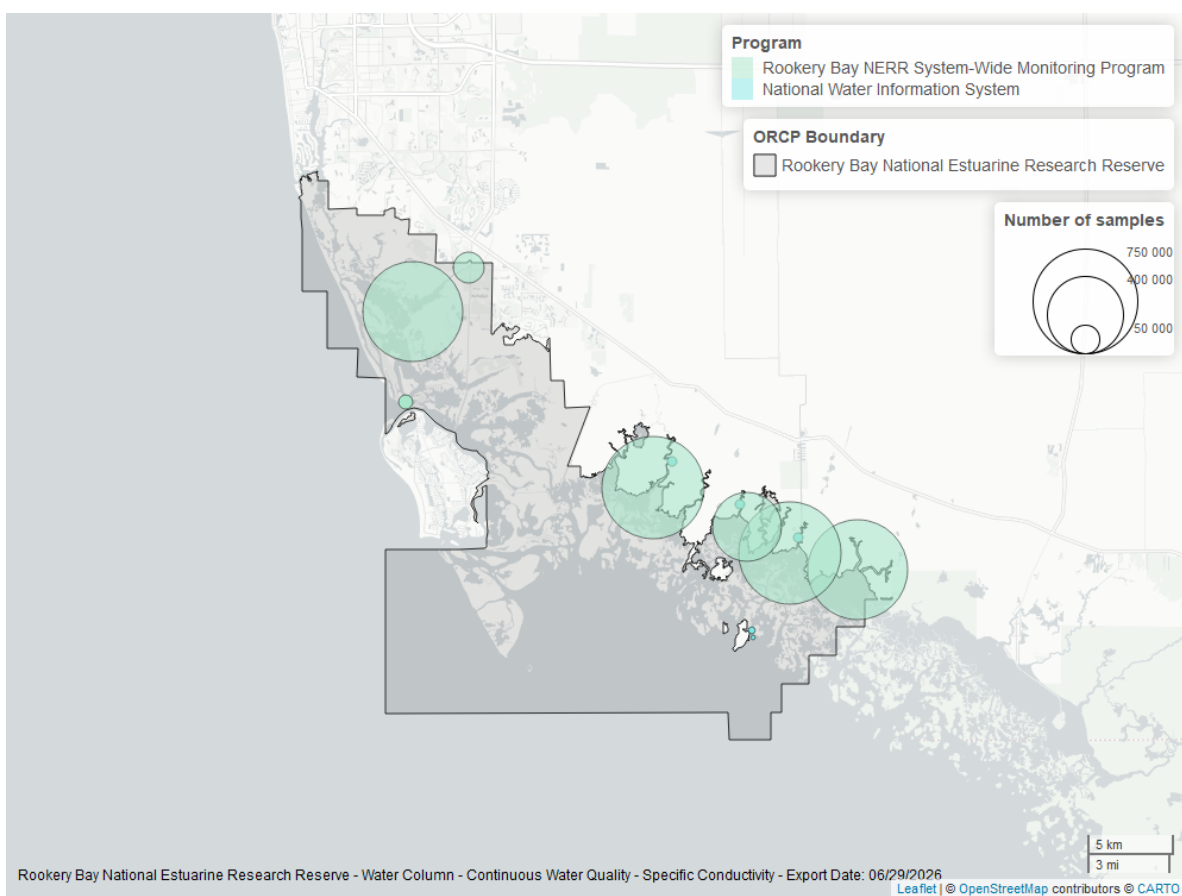


Figure 32: Map showing location of specific conductivity continuous water quality sampling locations within the boundaries of *Rookery Bay National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Clarity

Turbidity - Continuous

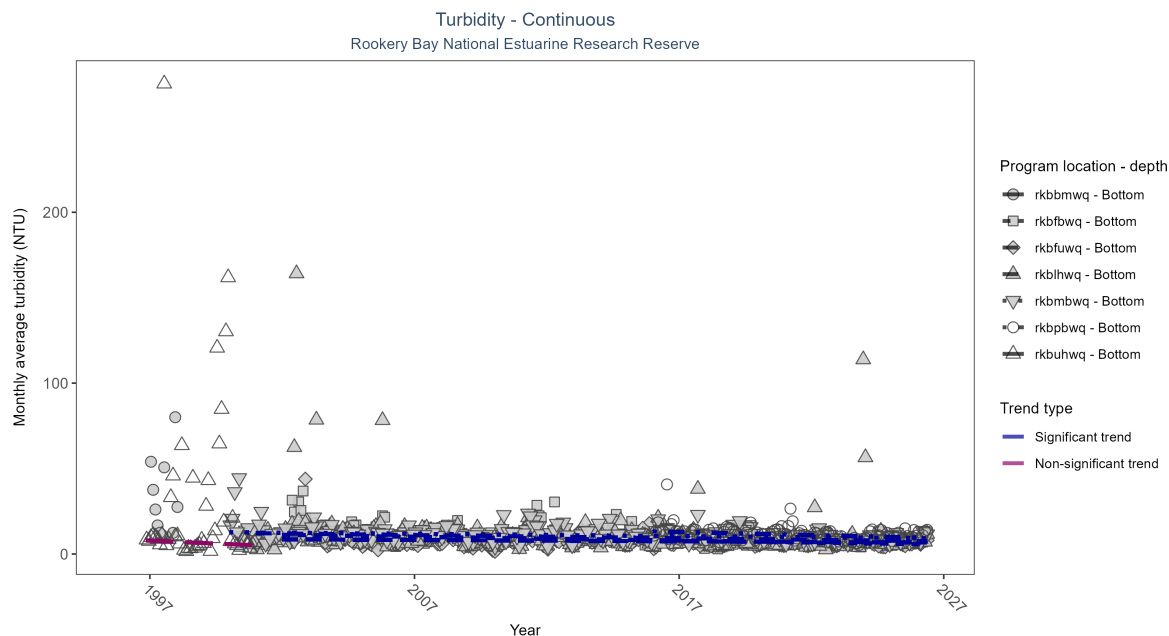


Figure 33: Scatter plot of monthly average turbidity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 16: At five program locations, monthly average turbidity decreased between 0.08 and 0.36 NTU per year. No detectable change in monthly average turbidity was observed at one location. There was insufficient data to fit a model for one location.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
rkbbmwq	Insufficient data	10654	2	1997 - 1998	11	-	-	-	-
rkbfwq	Decreasing trend	691451	25	2002 - 2026	7	-0.39	11.21	-0.23	0
rkbfuwq	Decreasing trend	664104	25	2002 - 2026	6	-0.19	8.57	-0.08	0
rkblhwq	Decreasing trend	657127	26	2001 - 2026	8	-0.27	12.18	-0.18	0
rkmbmwq	Decreasing trend	722768	27	2000 - 2026	9	-0.23	12.72	-0.13	0
rkbpbwq	Decreasing trend	339797	11	2016 - 2026	10	-0.29	13.23	-0.36	0
rkbuwq	No detectable trend	61608	5	1996 - 2000	5	-0.11	8.57	-0.7	0.52

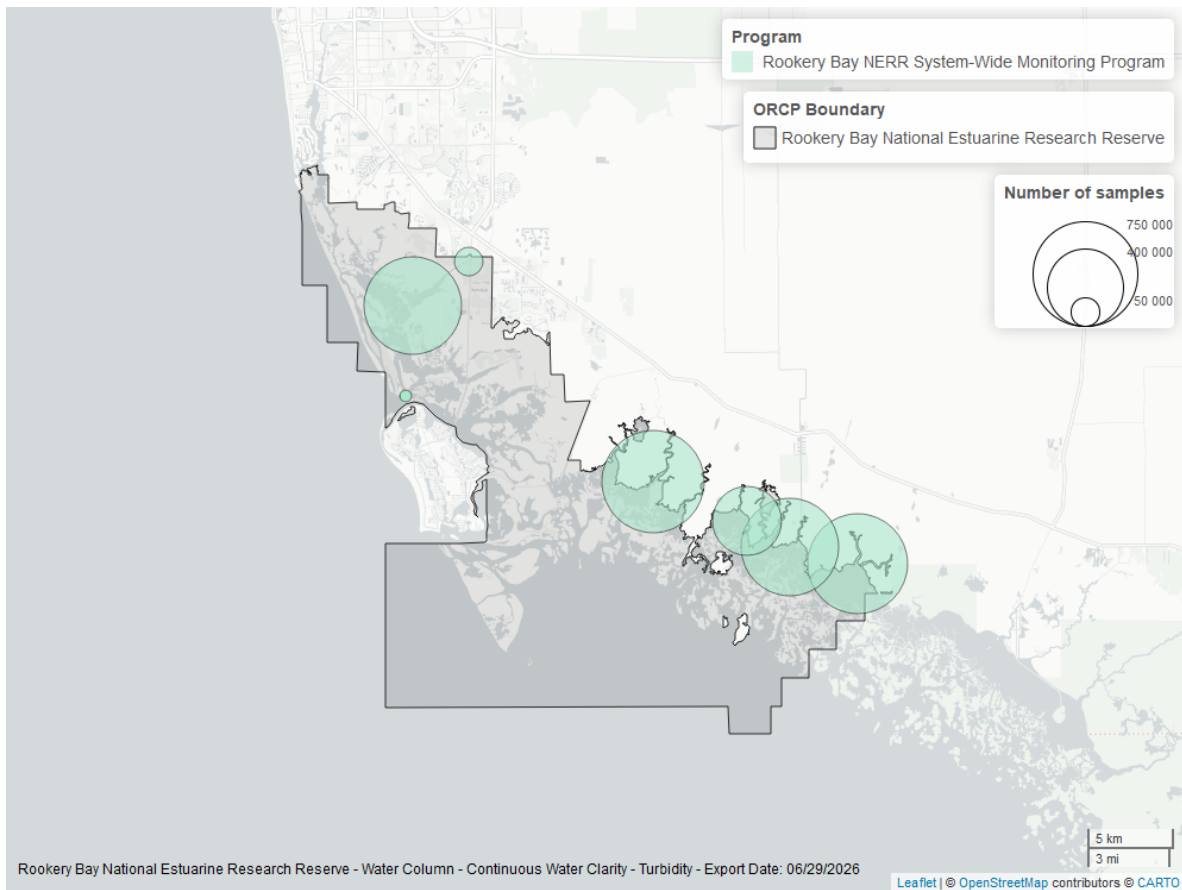


Figure 34: Map showing location of turbidity continuous water quality sampling locations within the boundaries of *Rookery Bay National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

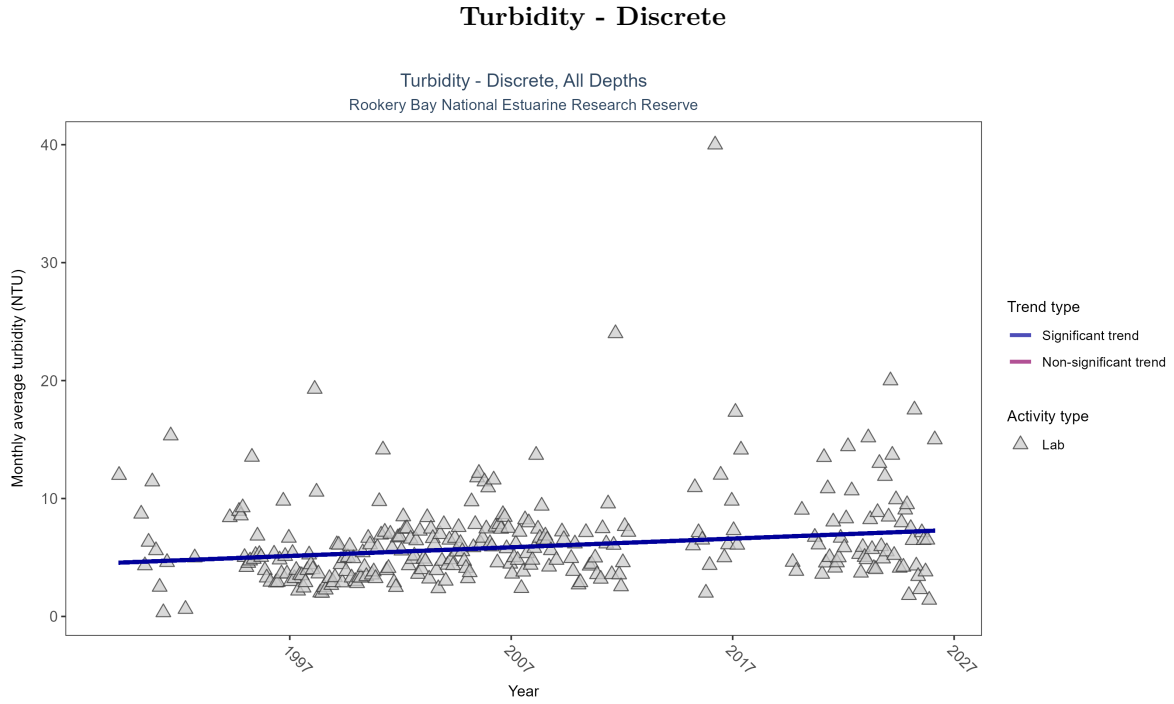


Figure 35: Scatter plot of monthly average turbidity over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only turbidity values measured in the laboratory (triangles) are included in the plot.

Table 17: Monthly average turbidity increased by 0.07 NTU per year, indicating a decrease in water clarity.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Increasing trend	6450	34	1989 - 2026	4.3825	0.17937	4.53913	0.07361	0

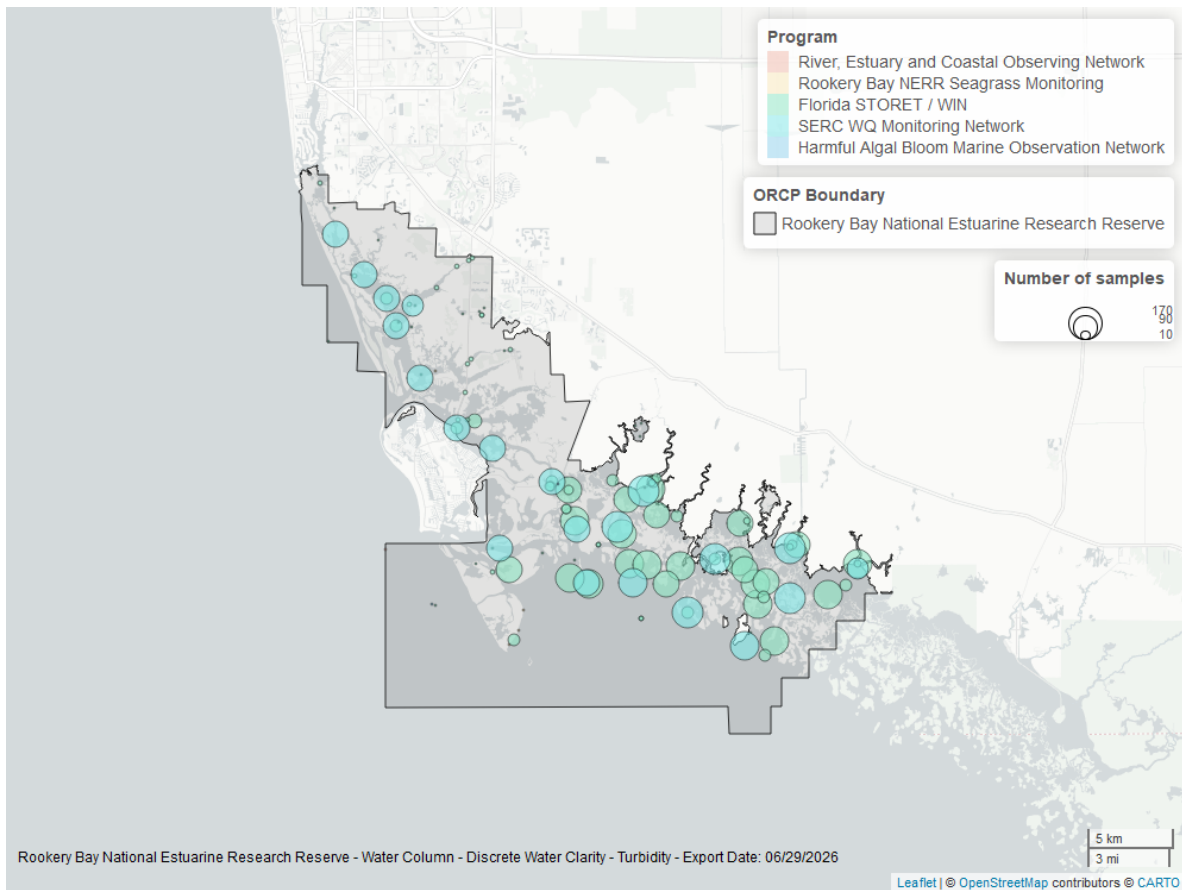


Figure 36: Map showing location of discrete water quality sampling locations within the boundaries of *Rookery Bay National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Total Suspended Solids - Discrete

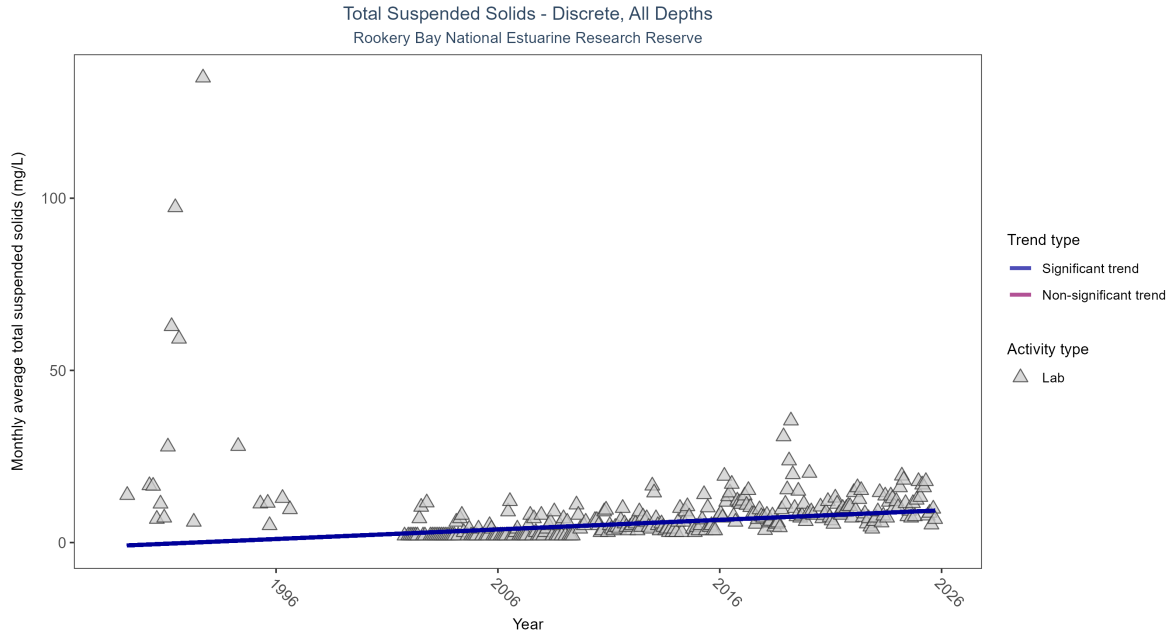


Figure 37: Scatter plot of monthly average total suspended solids (TSS) over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only TSS values obtained from laboratory analyses (triangles) are included in the plot.

Table 18: Monthly average total suspended solids increased by 0.28 mg/L per year, indicating a decrease in water clarity.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Increasing trend	926	32	1989 - 2025	8	0.38338	-0.91416	0.27778	0

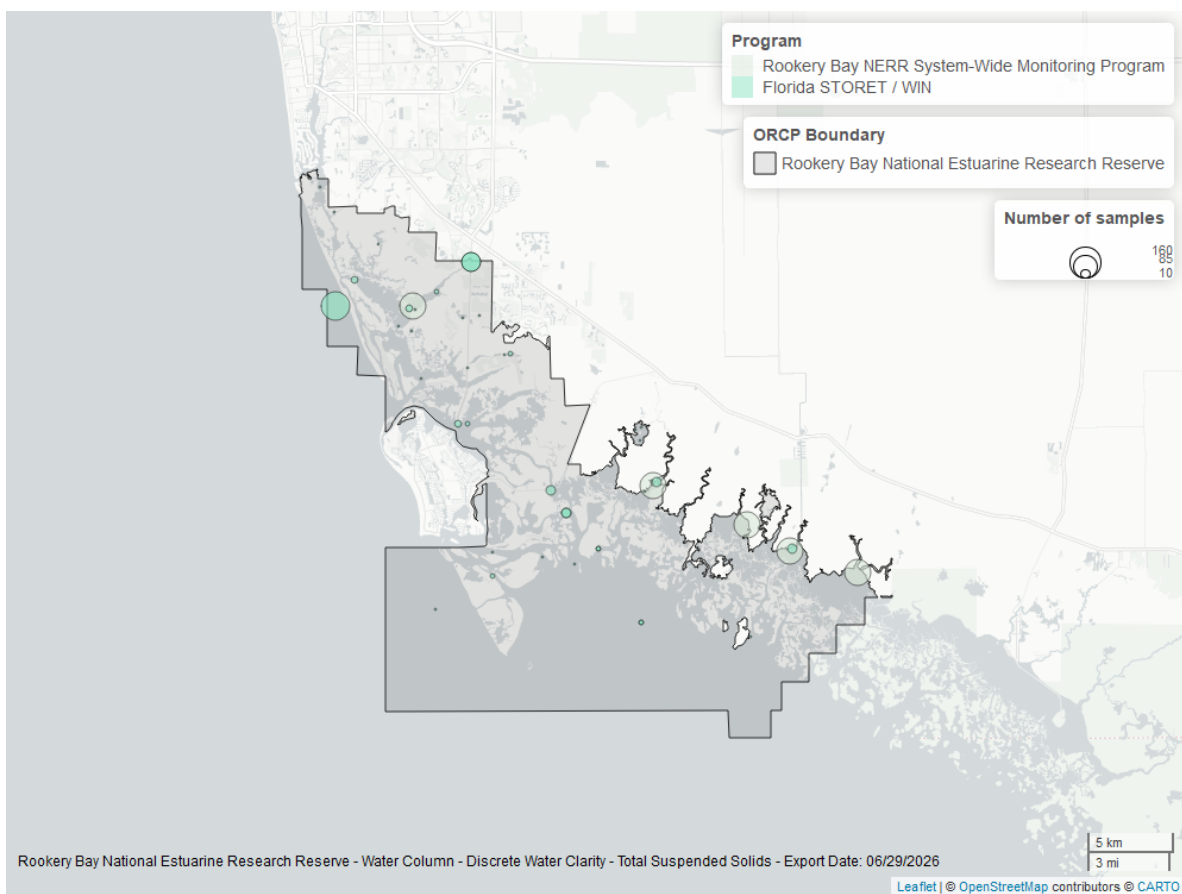


Figure 38: Map showing location of discrete water quality sampling locations within the boundaries of *Rookery Bay National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Chlorophyll a, Uncorrected for Pheophytin - Discrete

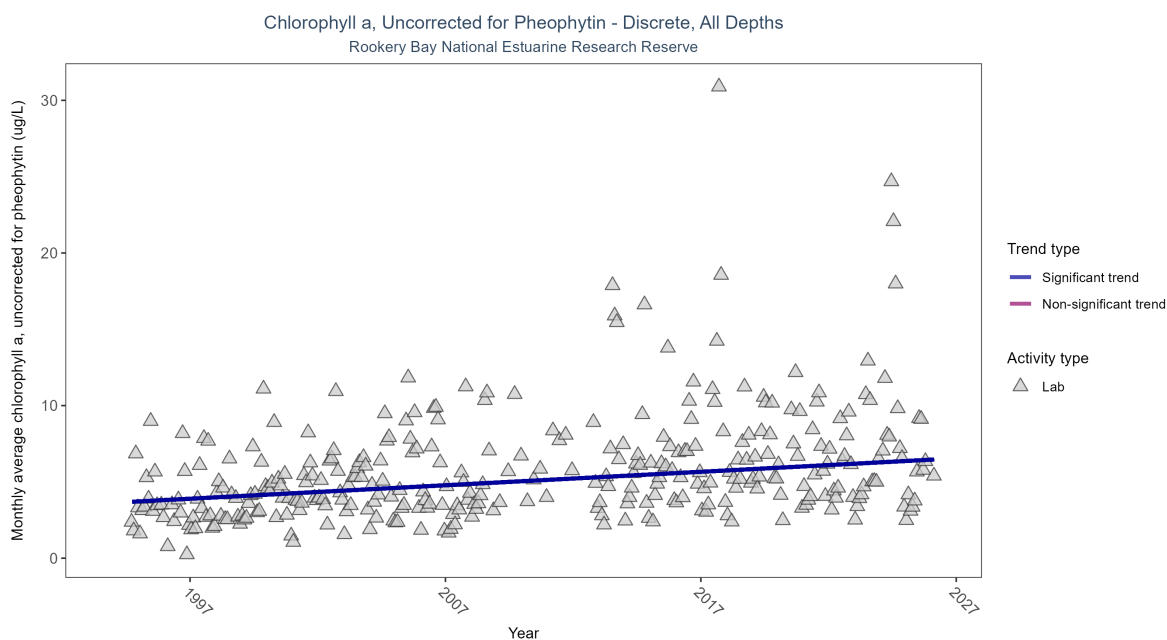


Figure 39: Scatter plot of monthly average levels of chlorophyll a, uncorrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 19: Monthly average chlorophyll a, uncorrected for pheophytin, increased by 0.09 ug/L per year, indicating a decrease in water clarity.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Increasing trend	5540	33	1994 - 2026	4.43848	0.3138	3.63368	0.08817	0

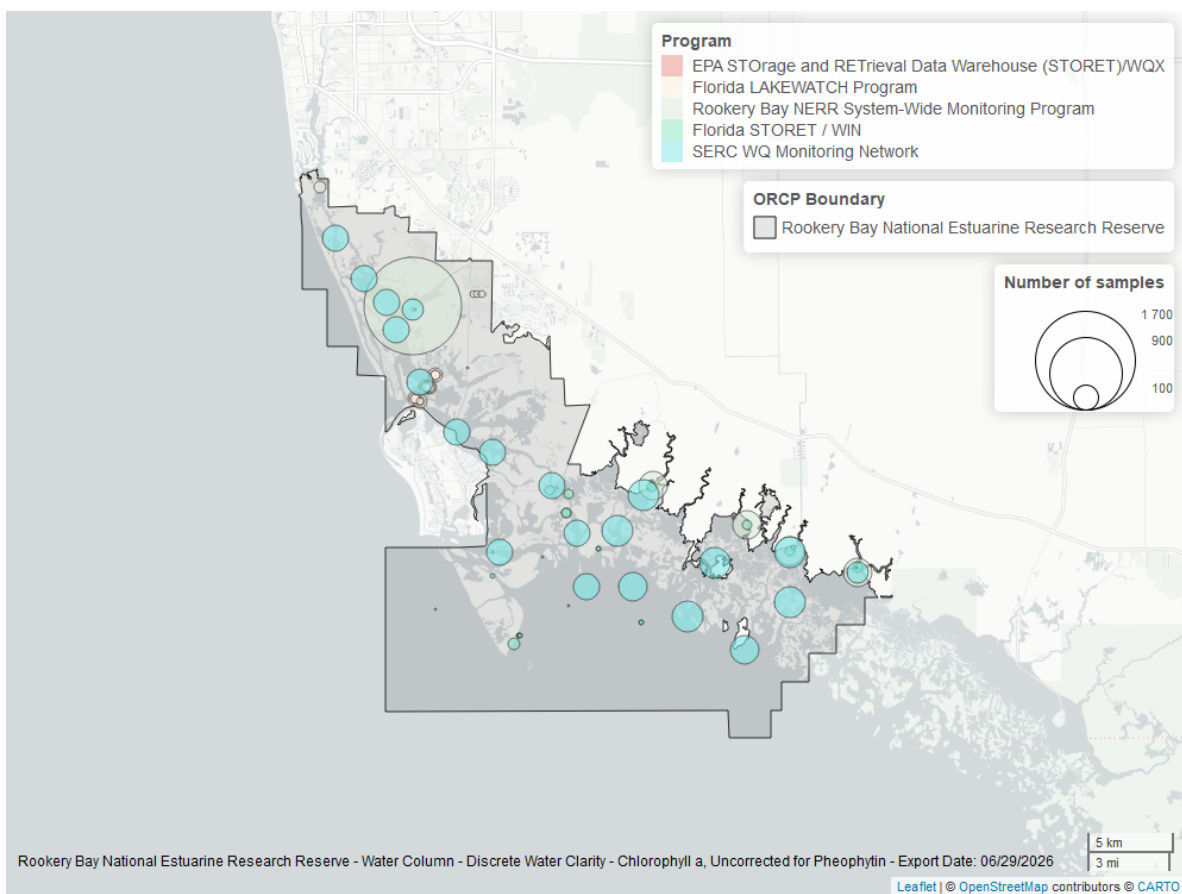


Figure 40: Map showing location of discrete water quality sampling locations within the boundaries of *Rookery Bay National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Chlorophyll a, Corrected for Pheophytin - Discrete

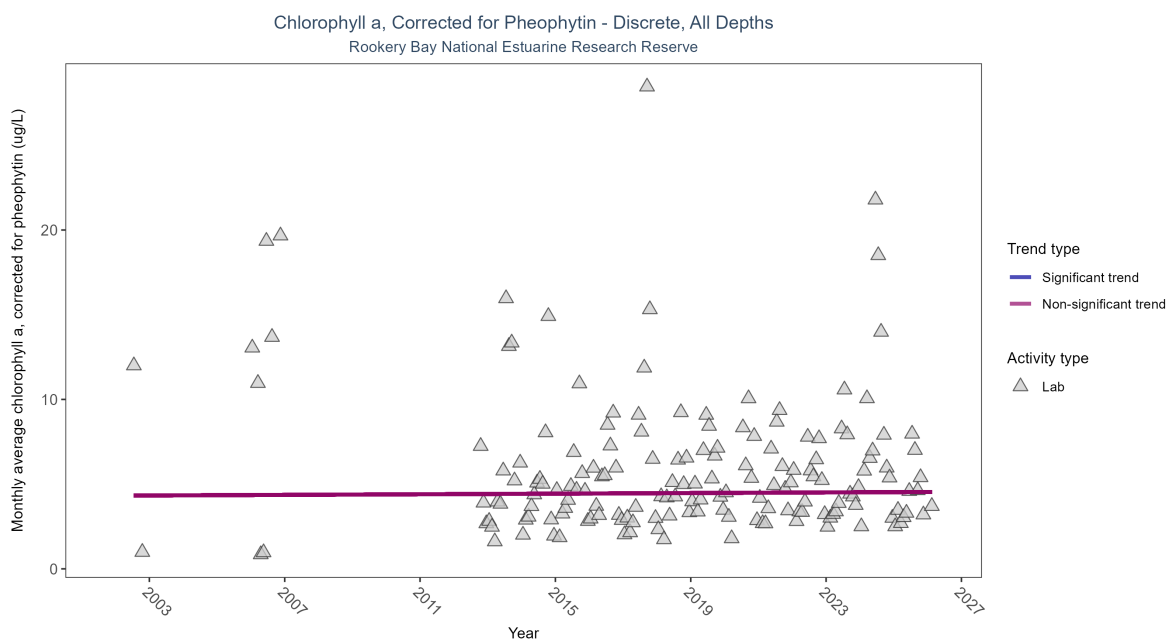


Figure 41: Scatter plot of monthly average levels of chlorophyll a, corrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 20: Chlorophyll a, corrected for pheophytin, showed no detectable trend between 2002 and 2026.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	2455	17	2002 - 2026	4.1	0.02082	4.32007	0.00905	0.7317

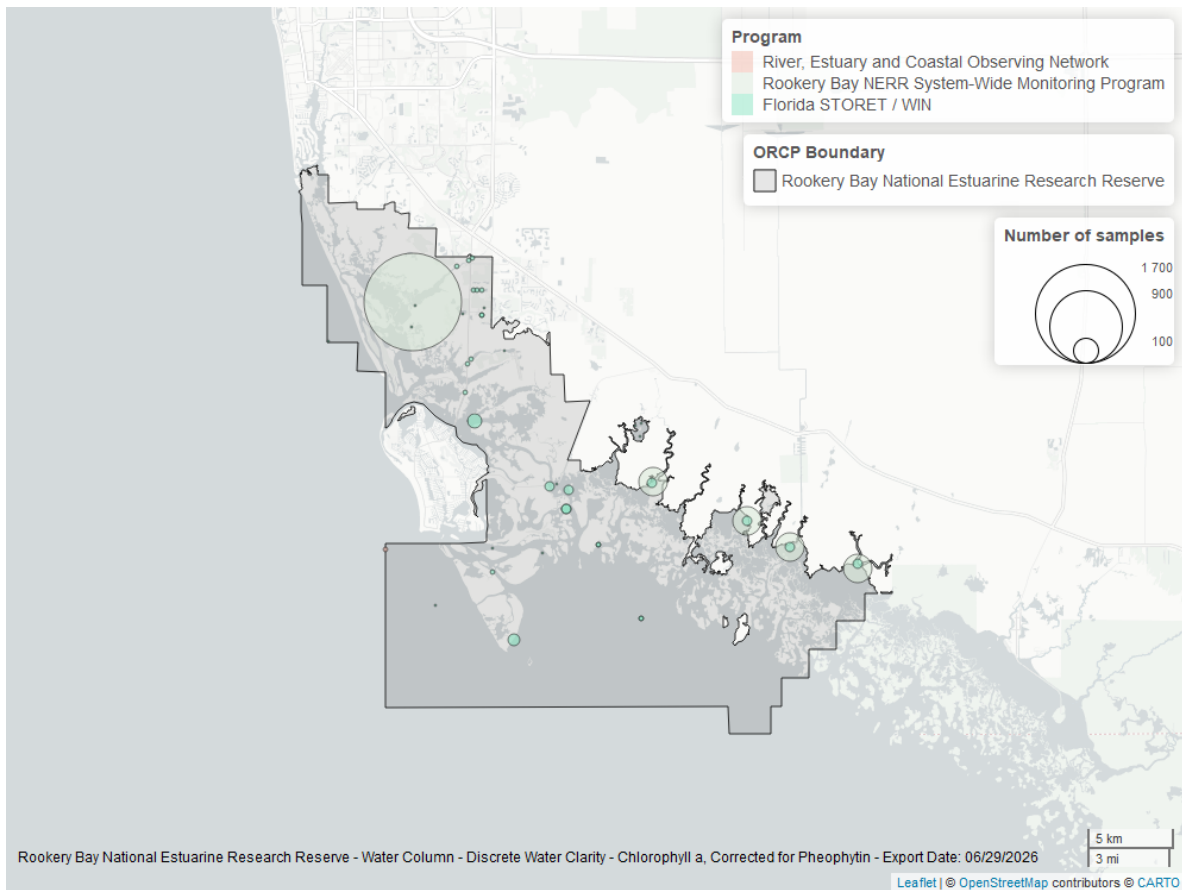


Figure 42: Map showing location of discrete water quality sampling locations within the boundaries of *Rookery Bay National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

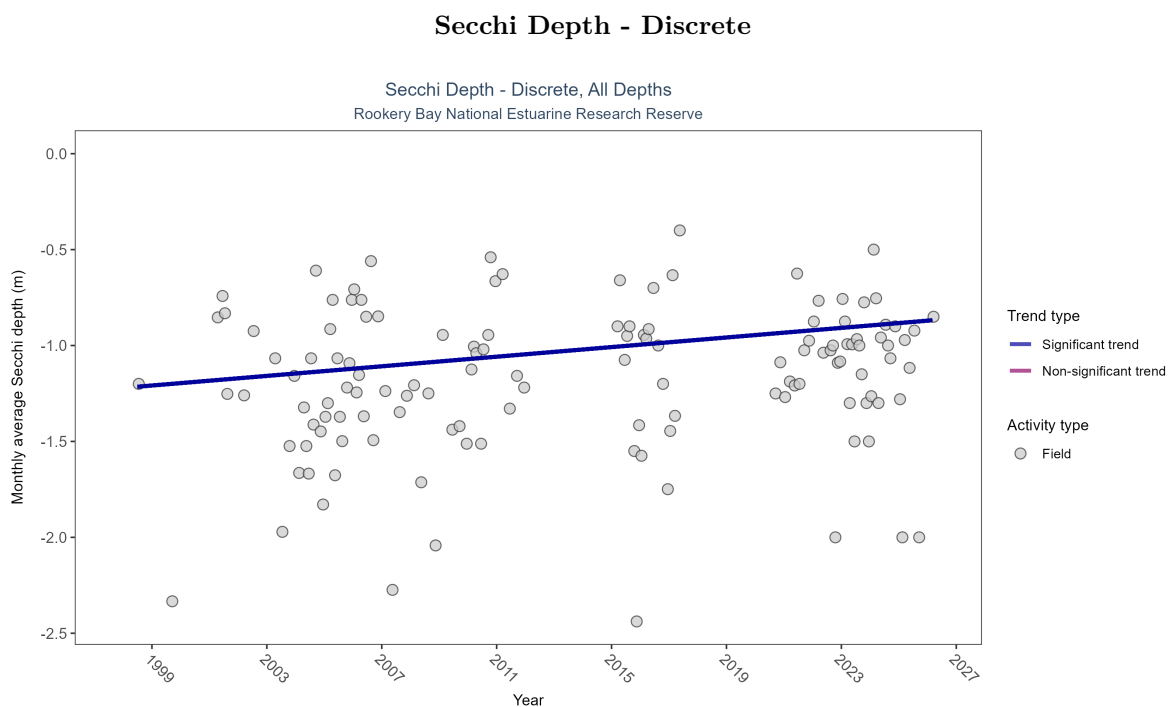


Figure 43: Scatter plot of monthly average Secchi depth over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Secchi depth is only measured in the field (circles).

Table 21: Monthly average Secchi depth became shallower by 0.01 m per year, indicating a decrease in water clarity.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Increasing trend	748	23	1998 - 2026	-1.06681	0.20304	-1.22103	0.01252	0.0018

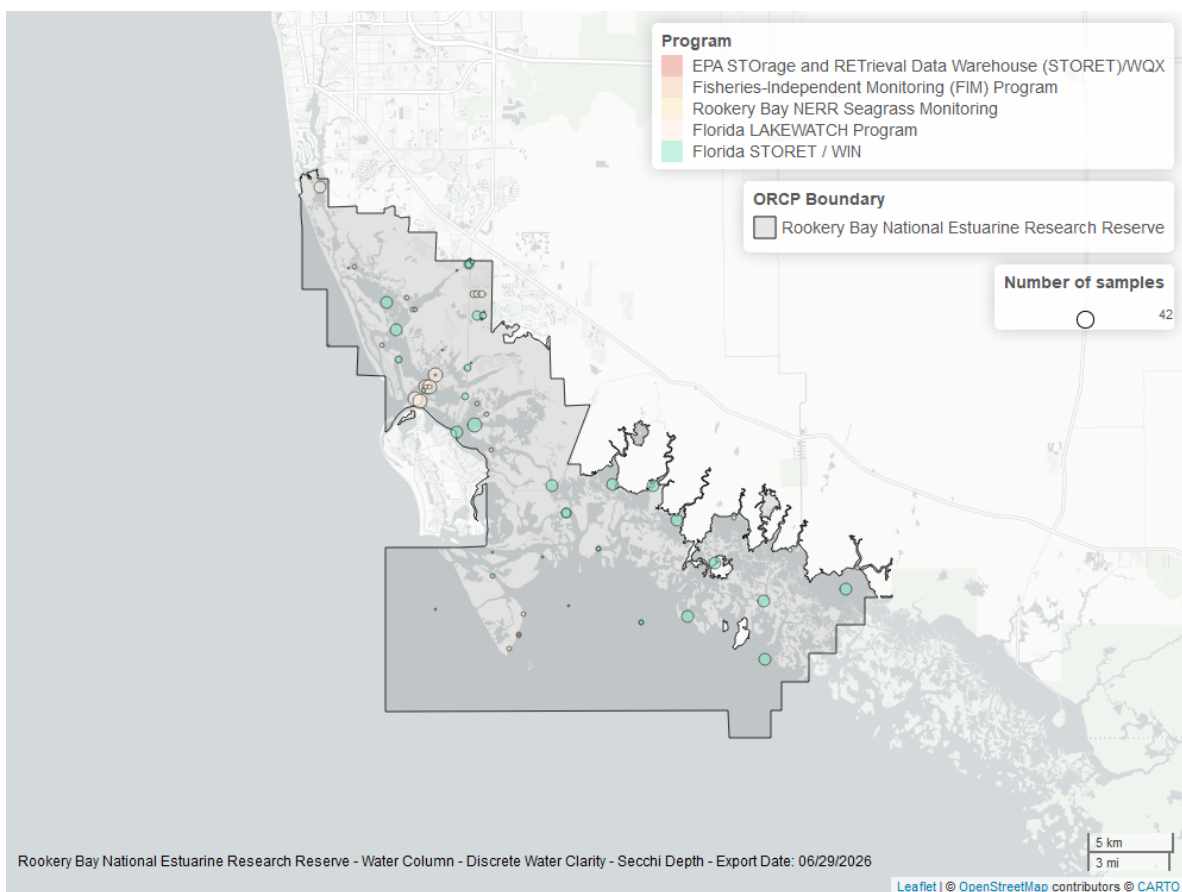


Figure 44: Map showing location of discrete water quality sampling locations within the boundaries of *Rookery Bay National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Colored Dissolved Organic Matter - Discrete

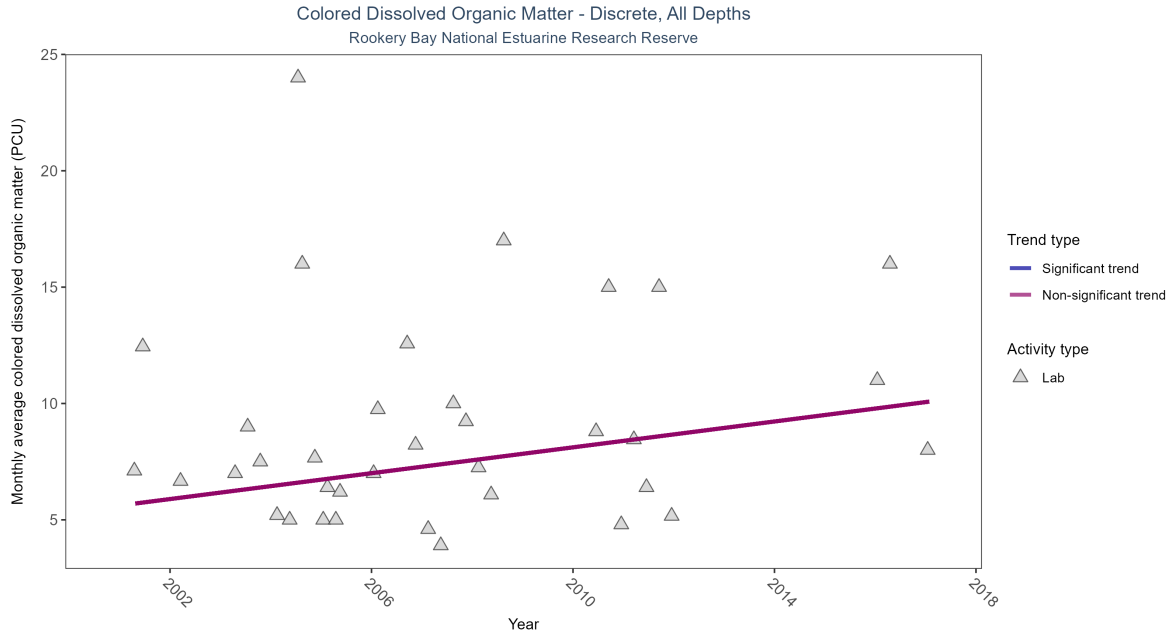


Figure 45: Scatter plot of monthly average colored dissolved organic matter (CDOM) over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed CDOM (triangles) is included in the plot.

Table 22: Colored dissolved organic matter showed no detectable trend between 2001 and 2017.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	193	12	2001 - 2017	8	0.3619	5.61538	0.27778	0.1533

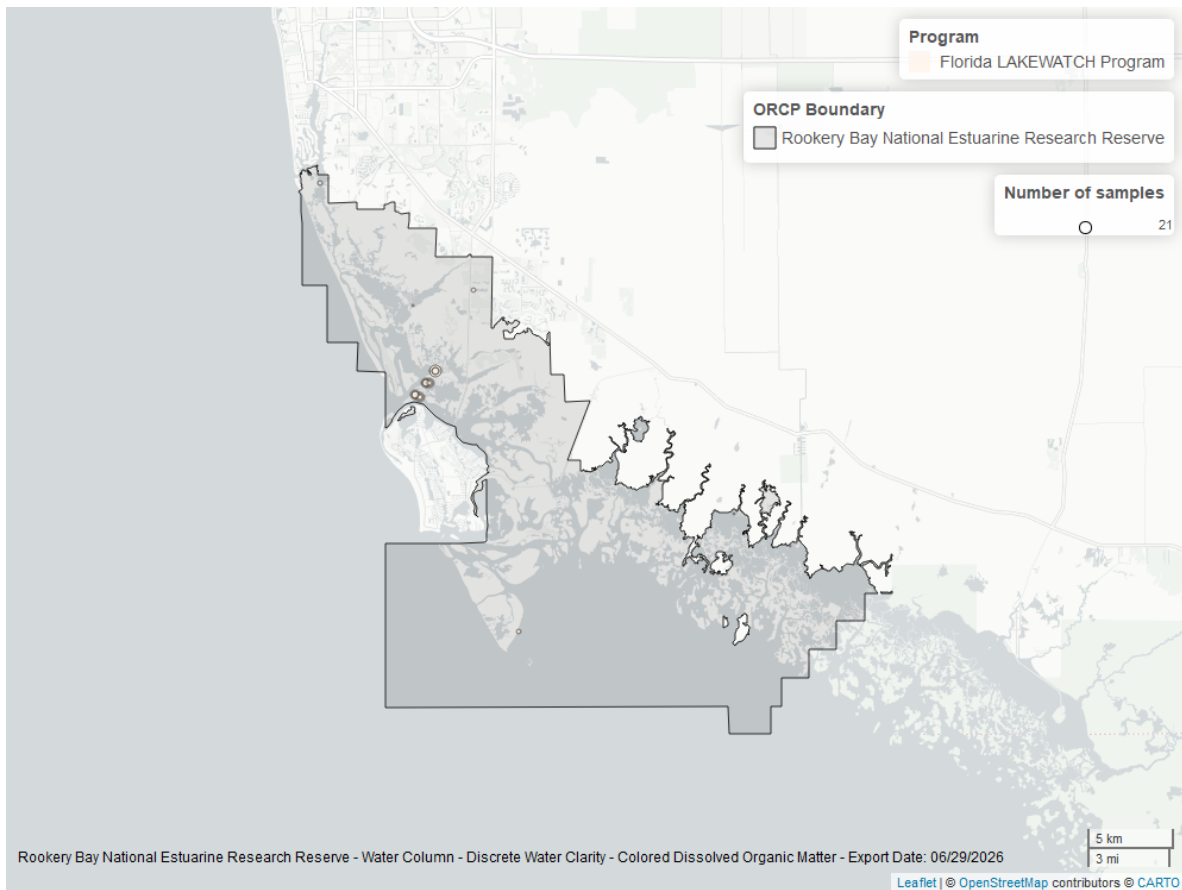


Figure 46: Map showing location of discrete water quality sampling locations within the boundaries of *Rookery Bay National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.